

[GETNET] Install Oracle RAC 19c in Linux – TRANS

Premissa da atividade: Instalar um RAC Linux, virtualizado no VMWare usando um deploy inicial, de máquinas StandAlone no Ansible.

Vantagens de usar o Ansible: Os hosts já saem configurados com todos os padrões da Getnet. Bastamos apenas remover os binários e dropar diskgroups e arquivos de configuração.

Deploy de 3 servidores StandAlone

Configuração dos IPs

NOTA: Importe solicitar ao Moises Von Scharten Pereira Roth voltar as máquinas criadas, do status 3 para status 1. Assim é possível reservar os demais IPs. No SSO, reservar os IPs do Interconnect, VIP e Scan (pré-requisito para RAC Oracle).

Configuração de IPs

GNCASHTL08920

IP Público: 180.104.52.47
IP Branca: 22.248.151.33
IP Backup: 22.240.151.33
Interconnect: 172.27.52.14
IP VIP: 180.104.52.51

GNCASHTL08921

IP Público: 180.104.52.48
IP Branca: 22.248.151.34
IP Backup: 22.240.151.34
Interconnect: 172.27.52.15
IP VIP 180.104.52.52

GNCASHTL08922

IP Público: 180.104.52.49
IP Branca: 22.248.151.35
IP Backup: 22.240.151.35
Interconnect: 172.27.52.16
IP VIP 180.104.52.53

DNS: vip-taponphone-db-oracle.getnet.local

IPS:

180.104.52.54
180.104.52.55
180.104.52.56

Imagens da reserva de IPs:

GNCASHTL08920

Seguir o último template vigente do Ansible para Red Hat 8 e Oracle Linux 19. A ideia de usar a esteira é já ter os dois nós do RAC configurados com o padrão da Getnet tais como packages, limites e diretórios.

Info Maquina x Info Maquina x Info Maquina x +

← → ↻ https://getnet.com.br/sgui/default/maquina?id_maquina=8920

Você deve se autenticar nessa rede antes de poder acessar a internet. [Abrir página de acesso à rede](#)

Getnet® aa58177

Configurações Atuais

IP Maquina	180.104.52.47	Configurado ☒	Excluir ☐	ATUALIZAR
IP Maquina	180.104.52.51	Configurado ☒	Excluir ☐	ATUALIZAR
IP Maquina	180.104.52.54	Configurado ☒	Excluir ☐	ATUALIZAR
IP Maquina	180.104.52.55	Configurado ☒	Excluir ☐	ATUALIZAR
IP Maquina	180.104.52.56	Configurado ☒	Excluir ☐	ATUALIZAR
IP Branca	22.248.151.33	Configurado ☒	Excluir ☐	ATUALIZAR
IP Backup	22.240.151.33	Configurado ☒	Excluir ☐	ATUALIZAR
IP INTERCONNECTORABLE	172.27.52.14	Configurado ☒	Excluir ☐	ATUALIZAR

Incluir Nova Configuração

IP INTERCONNECTORABLE **INCLUIR NOVO**

CONFIGURAÇÕES DE DNS

INFORMAÇÕES DO BIG-IP

INFORMAÇÕES DE NAT

GNCASHTL08921

Info Maquina x Info Maquina x Info Maquina x +

← → ↻ https://getnet.com.br/sgui/default/maquina?id_maquina=8921

Você deve se autenticar nessa rede antes de poder acessar a internet. [Abrir página de acesso à rede](#)

Getnet® aa58177

VLAN Serviço de dados Oracle RAC

SERVICE-NOW

CONFIGURAÇÕES DE REDE

Configurações Atuais

IP Maquina	180.104.52.48	Configurado ☒	Excluir ☐	ATUALIZAR
IP Maquina	180.104.52.52	Configurado ☒	Excluir ☐	ATUALIZAR
IP Branca	22.248.151.34	Configurado ☒	Excluir ☐	ATUALIZAR
IP Backup	22.240.151.34	Configurado ☒	Excluir ☐	ATUALIZAR
IP INTERCONNECTORABLE	172.27.52.15	Configurado ☒	Excluir ☐	ATUALIZAR

Incluir Nova Configuração

IP INTERCONNECTORABLE **INCLUIR NOVO**

CONFIGURAÇÕES DE DNS

INFORMAÇÕES DO BIG-IP

INFORMAÇÕES DE NAT

GNCASHTL08922

Info Maquina x Info Maquina x Info Maquina x +

← → ↻ https://getnet.com.br/sgui/default/maquina?id_maquina=8922

Você deve se autenticar nessa rede antes de poder acessar a internet. [Abrir página de acesso à rede](#)

Getnet® aa58177

SERVICE-NOW

CONFIGURAÇÕES DE REDE

Configurações Atuais

IP Maquina	180.104.52.49	Configurado ☒	Excluir ☐	ATUALIZAR
IP Maquina	180.104.52.53	Configurado ☒	Excluir ☐	ATUALIZAR
IP Branca	22.248.151.35	Configurado ☒	Excluir ☐	ATUALIZAR
IP Backup	22.240.151.35	Configurado ☒	Excluir ☐	ATUALIZAR
IP INTERCONNECTORABLE	172.27.52.16	Configurado ☒	Excluir ☐	ATUALIZAR

Incluir Nova Configuração

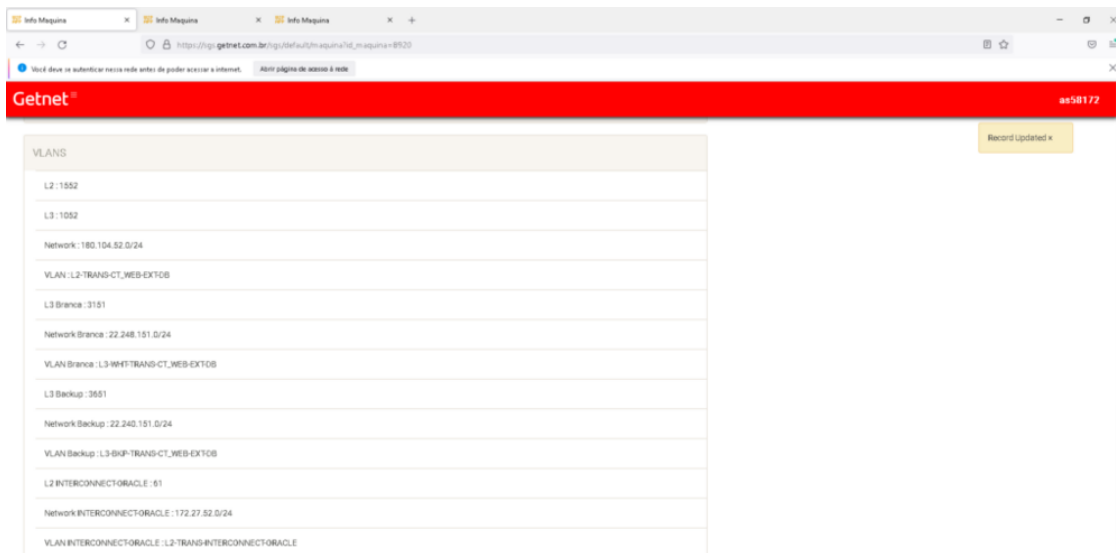
IP INTERCONNECTORABLE **INCLUIR NOVO**

CONFIGURAÇÕES DE DNS

INFORMAÇÕES DO BIG-IP

INFORMAÇÕES DE NAT

Ao fim da página, há um registro com todas as VLANs, lá podemos pegar o nome da VLAN do Interconnect:



Registro de DNS

Em exceção, não foi feito o registro do DNS durante a instalação do RAC pois o projeto já usava o nome vip-taponphone-db-oracle.getnet.local e só faremos o registro, quando forem migrar o ambiente.

Configuração do /etc/hosts (provisório a parte do DNS do Scan)

```
### Install RAC Oracle
180.104.52.47 GNCASHTL08920 GNCASHTL08920.getnet.local
22.240.151.33 GNCASHTL08920.bkp GNCASHTL08920.bkp.getnet.local
22.248.151.33 GNCASHTL08920.wt GNCASHTL08920.wt.getnet.local
180.104.52.51 GNCASHTL08920-vip GNCASHTL08920-vip.getnet.local

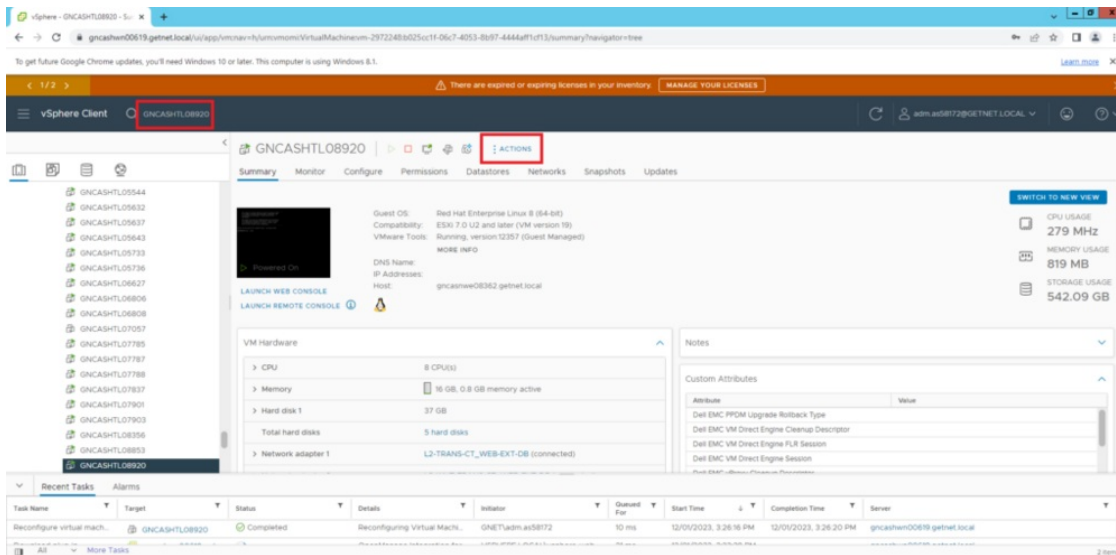
180.104.52.48 GNCASHTL08921 GNCASHTL08921.getnet.local
22.248.151.34 GNCASHTL08921.bkp GNCASHTL08921.bkp.getnet.local
22.248.151.34 GNCASHTL08921.wt GNCASHTL08921.wt.getnet.local
180.104.52.52 GNCASHTL08921-vip GNCASHTL08921-vip.getnet.local

180.104.52.49 GNCASHTL08922 GNCASHTL08922.getnet.local
22.240.151.35 GNCASHTL08922.bkp GNCASHTL08922.bkp.getnet.local
2.248.151.35 GNCASHTL08922.wt GNCASHTL08922.wt.getnet.local
180.104.52.53 GNCASHTL08922-vip GNCASHTL08922-vip.getnet.local

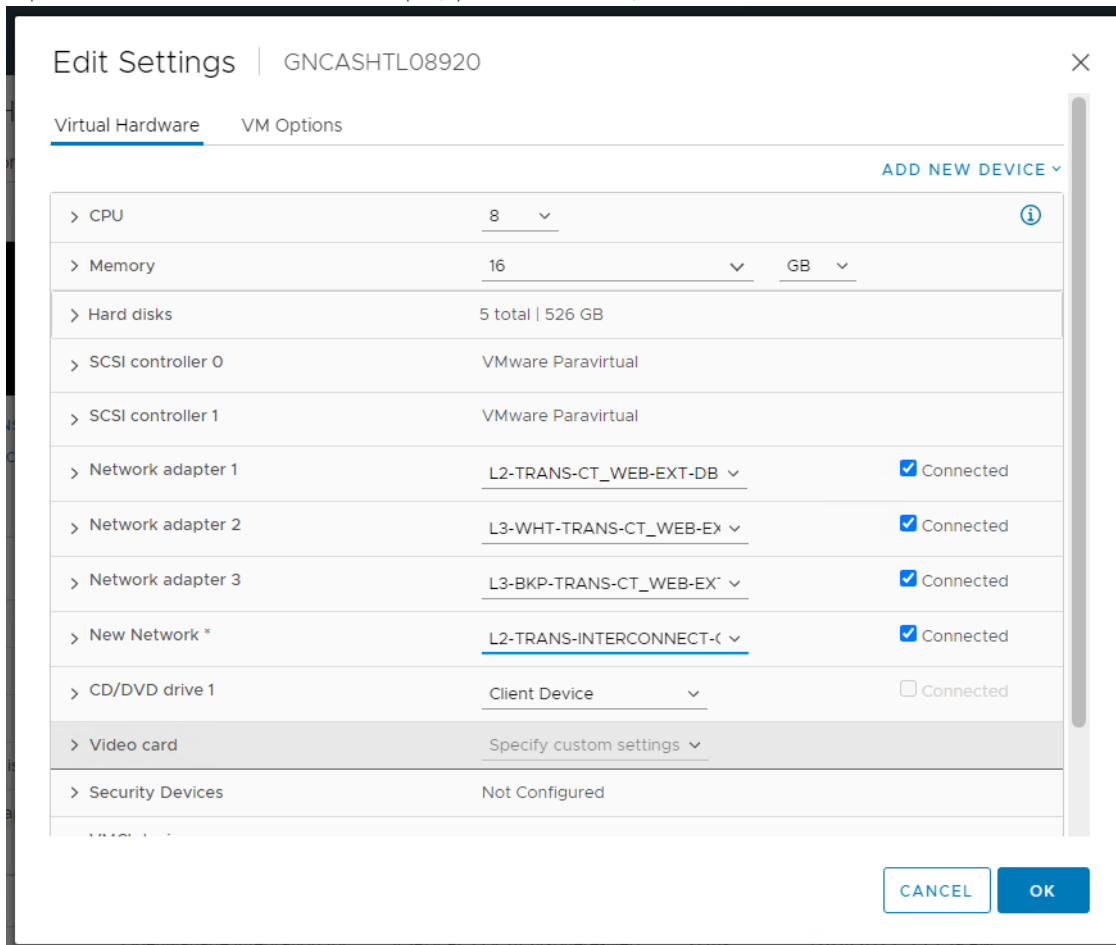
### Scan RAC Oracle
180.104.52.54 vip-taponphone-db-oracle vip-taponphone-db-oracle.getnet.local
180.104.52.55 vip-taponphone-db-oracle vip-taponphone-db-oracle.getnet.local
180.104.52.56 vip-taponphone-db-oracle vip-taponphone-db-oracle.getnet.local
```

Adição da interface de rede do Interconnect

NOTA: Adicionar a Interface de rede para as 3 VMs, vamos registrar apenas da primeira.
Entre na VM, vá em "Actions" e clique em "Edit Settings...":



Clique em "ADD NEW DEVICE" e escolha "Network Adapter", aparecerá uma nova linha, com nome de "New Network":



Na linha "New Network" escolha a seta pra baixo, afim de escolhermos a Vlan correta.
Escolha a VLAN indicada no SSO, no caso, L2-TRANS-INTERCONNECT-ORACLE:

Select Network

×

▼ oracle

Name	NSX Port Group ID	Distributed Switch
 L2-PROD-INTERCO...	--	DSwitch_PROD_Servico
 L2-TRANS-INTERC...	—	DSwitch_PROD_Servico

2 items

CANCEL

OK

Configuração das interfaces de rede, do Interconnect, a nível de S.O.

```
cd /etc/sysconfig/network-scripts/
cat ifcfg-ens193

GNCASHTL08920
# Generated by VMWare customization engine.
NAME=ens193
DEVICE=ens193
HWADDR=00:50:56:b6:3f:95
ONBOOT=yes
USERCTL=no
BOOTPROTO=static
NETMASK=255.255.248.0
IPADDR=172.27.52.14
PEERDNS=no
IPV6INIT=no
check_link_down() {
    return 1;
}

GNCASHTL08921
# Generated by VMWare customization engine.
NAME=ens193
DEVICE=ens193
HWADDR=00:50:56:b6:99:4a
ONBOOT=yes
USERCTL=no
BOOTPROTO=static
NETMASK=255.255.248.0
IPADDR=172.27.52.15
PEERDNS=no
IPV6INIT=no
check_link_down() {
    return 1;
}

GNCASHTL08922
# Generated by VMWare customization engine.
NAME=ens193
DEVICE=ens193
HWADDR=00:50:56:b6:6b:97
ONBOOT=yes
USERCTL=no
BOOTPROTO=static
NETMASK=255.255.248.0
IPADDR=172.27.52.16
PEERDNS=no
IPV6INIT=no
check_link_down() {
    return 1;
}
```

Ajustes de Packages faltantes

Pacotes X11 para usarmos a instalação gráfica.
NOTA: fiz nos 2 hosts, registrei apenas um deles:

```
[root@GNCASHTL08920 network-scripts]# yum install xorg-x11-xauth
Updating Subscription Management repositories.
Zabbix - Rhel6 x86_64                61 kB/s | 1.5 kB    00:00
Zabbix - Rhel8                      59 kB/s | 1.5 kB    00:00
Zabbix 6.0 RHEL7                    57 kB/s | 1.5 kB    00:00
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs)  95 kB/s | 4.5 kB    00:00
Zabbix - Rhel7                      59 kB/s | 1.5 kB    00:00
Red Hat Satellite Capsule 6.13 for RHEL 8 x86_64 (RPMs)  98 kB/s | 4.5 kB    00:00
Zabbix 6.0                          34 kB/s | 1.5 kB    00:00
Red Hat Satellite Maintenance 6.13 for RHEL 8 x86_64 (RPMs) 109 kB/s | 4.2 kB    00:00
Grafana                             42 kB/s | 1.5 kB    00:00
```

```
Hed Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs)            81 kB/s | 4.1 kB  00:00
Red Hat Satellite 6.13 for RHEL 8 x86_64 (RPMs)              108 kB/s | 4.5 kB  00:00
Red Hat Satellite Client 6 for RHEL 8 x86_64 (RPMs)           74 kB/s | 3.8 kB  00:00
MongoDB 5 EL7                  11 kB/s | 1.5 kB  00:00
Zabbix - Rhel6 I386             13 kB/s | 1.5 kB  00:00
Red Hat Satellite Utils 6.13 for RHEL 8 x86_64 (RPMs)         121 kB/s | 4.2 kB  00:00
Zabbix 5.0 frontend              36 kB/s | 1.5 kB  00:00
Zabbix 4.0                       17 kB/s | 1.5 kB  00:00
Dependencies resolved.

=====
===
Package                Architecture  Version              Repository              Size
=====
Installing:
xorg-x11-xauth          x86_64      1:1.0.9-12.el8       rhel-8-for-x86_64-appstream-rpms 39 k

Transaction Summary
=====
===
Install 1 Package

Total download size: 39 k
Installed size: 63 k
Is this ok [y/N]: y
Downloading Packages:
xorg-x11-xauth-1.0.9-12.el8.x86_64.rpm                170 kB/s | 39 kB  00:00
-----
Total                                                    169 kB/s | 39 kB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing      :                               1/1
Installing     : xorg-x11-xauth-1:1.0.9-12.el8.x86_64                1/1
Running scriptlet: xorg-x11-xauth-1:1.0.9-12.el8.x86_64                1/1
Verifying      : xorg-x11-xauth-1:1.0.9-12.el8.x86_64                1/1
Installed products updated.

Installed:
xorg-x11-xauth-1:1.0.9-12.el8.x86_64

Complete!
[root@GNCASHTL08920 network-scripts]# yum install xterm
Updating Subscription Management repositories.
Zabbix - Rhel6 x86_64                15 kB/s | 1.5 kB  00:00
Zabbix - Rhel8                      24 kB/s | 1.5 kB  00:00
Zabbix 6.0 RHEL7                    25 kB/s | 1.5 kB  00:00
Red Hat Enterprise Linux 8 for x86_64 - AppStream (RPMs)      113 kB/s | 4.5 kB  00:00
Zabbix - Rhel7                     60 kB/s | 1.5 kB  00:00
Red Hat Satellite Capsule 6.13 for RHEL 8 x86_64 (RPMs)      132 kB/s | 4.5 kB  00:00
Zabbix 6.0                        9.6 kB/s | 1.5 kB  00:00
Red Hat Satellite Maintenance 6.13 for RHEL 8 x86_64 (RPMs)  119 kB/s | 4.2 kB  00:00
Grafana                          60 kB/s | 1.5 kB  00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS (RPMs)        122 kB/s | 4.1 kB  00:00
Red Hat Satellite 6.13 for RHEL 8 x86_64 (RPMs)              56 kB/s | 4.5 kB  00:00
Red Hat Satellite Client 6 for RHEL 8 x86_64 (RPMs)           46 kB/s | 3.8 kB  00:00
MongoDB 5 EL7                  62 kB/s | 1.5 kB  00:00
Zabbix - Rhel6 I386             60 kB/s | 1.5 kB  00:00
Red Hat Satellite Utils 6.13 for RHEL 8 x86_64 (RPMs)         57 kB/s | 4.2 kB  00:00
Zabbix 5.0 frontend              22 kB/s | 1.5 kB  00:00
Zabbix 4.0                       53 kB/s | 1.5 kB  00:00
Dependencies resolved.

=====
===
Package                Architecture  Version              Repository              Size
=====
Installing:
xorg-x11-xauth          x86_64      1:1.0.9-12.el8       rhel-8-for-x86_64-appstream-rpms 39 k
```

```
xterm          x86_64      331-2.el8      rhel-8-for-x86_64-appstream-rpms      529 k
Installing dependencies:
xterm-resize    x86_64      331-2.el8      rhel-8-for-x86_64-appstream-rpms      39 k
Installing weak dependencies:
xorg-x11-fonts-misc    noarch      7.5-19.el8      rhel-8-for-x86_64-appstream-rpms      5.8 M

Transaction Summary
=====
===
Install 3 Packages

Total download size: 6.3 M
Installed size: 8.3 M
Is this ok [y/N]: y
Downloading Packages:
(1/3): xterm-resize-331-2.el8.x86_64.rpm                239 kB/s | 39 kB    00:00
(2/3): xterm-331-2.el8.x86_64.rpm                      1.2 MB/s | 529 kB   00:00
(3/3): xorg-x11-fonts-misc-7.5-19.el8.noarch.rpm        5.9 MB/s | 5.8 MB   00:00
-----
Total                                                    6.5 MB/s | 6.3 MB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing      :                                          1/1
Installing    : xterm-resize-331-2.el8.x86_64                1/3
Installing    : xorg-x11-fonts-misc-7.5-19.el8.noarch        2/3
Running scriptlet: xorg-x11-fonts-misc-7.5-19.el8.noarch      2/3
Installing    : xterm-331-2.el8.x86_64                      3/3
Running scriptlet: xterm-331-2.el8.x86_64                    3/3
Verifying     : xorg-x11-fonts-misc-7.5-19.el8.noarch        1/3
Verifying     : xterm-331-2.el8.x86_64                      2/3
Verifying     : xterm-resize-331-2.el8.x86_64                3/3
Installed products updated.

Installed:
xorg-x11-fonts-misc-7.5-19.el8.noarch    xterm-331-2.el8.x86_64    xterm-resize-331-2.el8.x86_64

Complete!
[root@GNCASHTL08920 network-scripts]#

[root@GNCASHTL08920 home]# mkdir adm.as58172
[root@GNCASHTL08920 home]# chown adm.as58172 adm.as58172
[root@GNCASHTL08920 home]# chmod 775 adm.as58172
```

Remoção dos binários de StandAlone

Baixe o banco, drop dos diskgroups, remova os binários do Oracle Database e Grid Infrastructure e limpe todos os arquivos de instalação.
NOTA: Fazer nos 3 nós, no caso da documentação, mostro apenas um dos nós.

Baixando o Banco:


```
[oracle@GNCASHTL08920 ~]$ . db_profile
[oracle@GNCASHTL08920 ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Fri Dec 1 15:37:15 2023
Version 19.21.0.0.0

Copyright (c) 1982, 2022, Oracle. All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0
[oracle@GNCASHTL08920 ~]$ exit
logout
```

Baixando o Grid:

```
[root@GNCASHTL08920 home]# su - grid
Last login: Fri Dec 1 13:13:34 -03 2023
[grid@GNCASHTL08920 ~]$ . grid_profile
[grid@GNCASHTL08920 ~]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncashtl08920'
CRS-2673: Attempting to stop 'ora.evmd' on 'gncashtl08920'
CRS-2673: Attempting to stop 'ora.ARCH_TAPONPHONE.dg' on 'gncashtl08920'
CRS-2673: Attempting to stop 'ora.DATA_AUX_TAPONPHONE.dg' on 'gncashtl08920'
CRS-2673: Attempting to stop 'ora.DATA_TAPONPHONE.dg' on 'gncashtl08920'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncashtl08920'
CRS-2677: Stop of 'ora.DATA_AUX_TAPONPHONE.dg' on 'gncashtl08920' succeeded
CRS-2677: Stop of 'ora.DATA_TAPONPHONE.dg' on 'gncashtl08920' succeeded
CRS-2677: Stop of 'ora.ARCH_TAPONPHONE.dg' on 'gncashtl08920' succeeded
CRS-2673: Attempting to stop 'ora.asm' on 'gncashtl08920'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncashtl08920' succeeded
CRS-2677: Stop of 'ora.evmd' on 'gncashtl08920' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncashtl08920' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncashtl08920'
CRS-2677: Stop of 'ora.cssd' on 'gncashtl08920' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncashtl08920' has completed
CRS-4133: Oracle High Availability Services has been stopped.
[grid@GNCASHTL08920 ~]$
```

Dropando os discos do ASM:

```

[root@GNCASHTL08920 home]# oracleasm listdisks
ARCH01_TAPONPHONE
ARCH02_TAPONPHONE
ARCH03_TAPONPHONE
ARCH04_TAPONPHONE
DATA01_AUX_TAPONPHONE
DATA01_TAPONPHONE
DATA02_AUX_TAPONPHONE
DATA02_TAPONPHONE
DATA03_AUX_TAPONPHONE
DATA03_TAPONPHONE
DATA04_AUX_TAPONPHONE
DATA04_TAPONPHONE

[root@GNCASHTL08920 home]# oracleasm deletedisk ARCH01_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk ARCH02_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk ARCH03_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk ARCH04_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA01_AUX_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA01_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA02_AUX_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA02_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA03_AUX_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA03_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA04_AUX_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]# oracleasm deletedisk DATA04_TAPONPHONE
Clearing disk header: done
Dropping disk: done
[root@GNCASHTL08920 home]#

[root@GNCASHTL08920 home]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@GNCASHTL08920 home]# oracleasm listdisks

```

Removendo a instalação de Banco e Grid:

NOTA: faça nos 3 nós

```

root@GNCASHTL08921 oracle]# cd /u01/
[root@GNCASHTL08921 u01]# rm -rf app

[root@GNCASHTL08920 home]# cd /tmp
[root@GNCASHTL08920 tmp]# rm -rf *
[root@GNCASHTL08920 tmp]# rm -rf /etc/orainst.loc
[root@GNCASHTL08920 tmp]# cd /etc/oracle
[root@GNCASHTL08920 oracle]# rm -rf *
[root@GNCASHTL08920 oracle]# rm -rf /etc/oratab
[root@GNCASHTL08920 oracle]# rm -rf /tmp/.oracle
[root@GNCASHTL08920 oracle]# /usr/lib/methods/ucfgacfsctl -l ofscctl
-bash: /usr/lib/methods/ucfgacfsctl: No such file or directory
[root@GNCASHTL08920 oracle]# /usr/lib/methods/ucfgadmctl -l advmctl
-bash: /usr/lib/methods/ucfgadmctl: No such file or directory
[root@GNCASHTL08920 oracle]# /usr/lib/methods/udefacfsctl -l ofscctl
-bash: /usr/lib/methods/udefacfsctl: No such file or directory
[root@GNCASHTL08920 oracle]# /usr/lib/methods/udefadmctl -l advmctl
-bash: /usr/lib/methods/udefadmctl: No such file or directory
[root@GNCASHTL08920 oracle]# /usr/sbin/rmauth -h oracle
-bash: /usr/sbin/rmauth: No such file or directory
[root@GNCASHTL08920 oracle]# rmrole oracle_devmgmt
-bash: rmrole: command not found
[root@GNCASHTL08920 oracle]# setkst
-bash: setkst: command not found
[root@GNCASHTL08920 oracle]# rm /usr/lib/drivers/oracle*
rm: cannot remove '/usr/lib/drivers/oracle*': No such file or directory
[root@GNCASHTL08920 oracle]#
[root@GNCASHTL08920 oracle]# rm /usr/lib/methods/*advm* /usr/lib/methods/*acfs*
rm: cannot remove '/usr/lib/methods/*advm*': No such file or directory
rm: cannot remove '/usr/lib/methods/*acfs*': No such file or directory
[root@GNCASHTL08920 oracle]# rm -rf /sbin/helpers/acfs
[root@GNCASHTL08920 oracle]# rm /usr/sbin/acfsutil* /usr/sbin/advmutil*
rm: cannot remove '/usr/sbin/acfsutil*': No such file or directory
rm: cannot remove '/usr/sbin/advmutil*': No such file or directory
[root@GNCASHTL08920 oracle]# rm /sbin/acfsutil* /sbin/advmutil*
rm: cannot remove '/sbin/acfsutil*': No such file or directory
rm: cannot remove '/sbin/advmutil*': No such file or directory
[root@GNCASHTL08920 oracle]#

```

Faça o Shutdown em todos os hosts e vamos configurar a parte dos discos:

```

[root@GNCASHTL08920 u01]# shutdown -h now
Connection to gncashtl08920 closed by remote host.
Connection to gncashtl08920 closed.

```

Compartilhamento de Discos

O para o nó principal, GNCASHTL08920, vamos manter os discos de ASM e vamos fazer a configuração de compartilhamento. Os outros 2 nós, GNCASHTL08921 e GNCASHTL08922, vamos remover os discos de ASM e adicionar os discos do nó 01.

NOTA: Na Getnet, atualmente os hosts já estão saindo com 2 controladoras e os discos de ASM estão alocados na segunda controladora, sendo assim, fica fácil a identificação. Veja que o Hard disk 1 e 2 estão na SCSI(0:x) e os discos de ASM estão na SCSI(1:x), precisamos apenas deletar os discos da SCSI(1:x):

Edit Settings
GNCASHTL08922

Virtual Hardware
VM Options

ADD NEW DEVICE

> CPU	8		
> Memory	16	GB	
> Hard disks	5 total 526 GB		
> Hard disk 1	37 GB SCSI(0:0)		
> Hard disk 2	162 GB SCSI(0:1)		
> Hard disk 3	145 GB SCSI(1:0)		
> Hard disk 4	145 GB SCSI(1:1)		
> Hard disk 5	37 GB SCSI(1:2)		
> SCSI controller 0	VMware Paravirtual		
> SCSI controller 1	VMware Paravirtual		
> Network adapter 1	L2-TRANS-CT_WEB-EXT-DB	☒	Connect...
> Network adapter 2	L3-WHT-TRANS-CT_WEB-EX	☒	Connect...
> Network adapter 3	L3-BKP-TRANS-CT_WEB-EX	☒	Connect...

CANCEL
OK

Marque a opção de deletar do datastore:
 LEMBRE-SE, repetir as ações de delete para GNCASHTL08921 e GNCASHTL08922.

Edit Settings
GNCASHTL08922

Virtual Hardware
VM Options

ADD NEW DEVICE

> CPU	8		
> Memory	16	GB	
> Hard disks	5 total 526 GB		
> Hard disk 1	37 GB SCSI(0:0)		
> Hard disk 2	162 GB SCSI(0:1)		
> Hard disk 3 *	Device will be removed	☒	Delete files from datastore ↵
> Hard disk 4 *	Device will be removed	☒	Delete files from datastore ↵
> Hard disk 5 *	Device will be removed	☒	Delete files from datastore ↵
> SCSI controller 0	VMware Paravirtual		
> SCSI controller 1	VMware Paravirtual		
> Network adapter 1	L2-TRANS-CT_WEB-EXT-DB	☒	Connect...
> Network adapter 2	L3-WHT-TRANS-CT_WEB-EX	☒	Connect...

CANCEL
OK

Agora, desabilitar o CTK nas 3 máquinas.
 Edit Settings > VM Options > Advanced > Edit Configuration:

Edit Settings

GNCASHTL08920

> Power management

Expand for power management settings

> VMware Tools

Expand for VMware Tools settings

> Boot Options

Expand for boot options

▼ Advanced

Settings

☐ Disable acceleration

☒ Enable logging

Debugging and statistics

Run normally

Swap file location

☒ Default

Use the settings of the cluster or host containing the virtual machine.

☐ Virtual machine directory

Store the swap files in the same directory as the virtual machine.

☐ Datastore specified by host

Store the swap files in the datastore specified by the host to be used for swap files. If not possible, store the swap files in the same directory as the virtual machine. Using a datastore that is not visible to both hosts during vMotion might affect the vMotion performance for the affected virtual machines.

Configuration Parameters

EDIT CONFIGURATION...

CANCEL

OK

Filtre pela string CTK e altere todos de TRUE para FALSE:

Edit Settings

GNCASHTL08920

> Power management

Expand for power management settings

Configuration Parameters

⚠ Modify or add configuration parameters as needed for experimental features or as instructed by technical support. Empty values will be removed (supported on ESXi 6.0 and later).

ADD CONFIGURATION PARAMS

Name	Value
ctkEnabled	FALSE
scsi0:0.ctkEnabled	FALSE
scsi0:1.ctkEnabled	FALSE
scsi1:0.ctkEnabled	FALSE
scsi1:1.ctkEnabled	FALSE
scsi1:2.ctkEnabled	FALSE

CANCEL

OK

CANCEL

OK

Outro ponto importante é mudar a SCSI Controller 1 dos 3 hosts para virtual:
 Change Type: VMware Paravirtual
 SCSI Bus Sharing: Virtual

Edit Settings
GNCASHTL08920

Virtual Hardware
VM Options

ADD NEW DEVICE

> CPU	8		
> Memory	16	GB	
> Hard disks	5 total 526 GB		
> SCSI controller 0	VMware Paravirtual		
> SCSI controller 1 *	VMware Paravirtual		
Change Type	VMware Paravirtual		
SCSI Bus Sharing	Virtual		
> Network adapter 1	L2-TRANS-CT_WEB-EXT-DB	☒	Connect...
> Network adapter 2	L3-WHT-TRANS-CT_WEB-EX	☒	Connect...
> Network adapter 3	L3-BKP-TRANS-CT_WEB-EX	☒	Connect...
> Network adapter 4	L2-TRANS-INTERCONNECT-C	☒	Connect...
> CD/DVD drive 1	Client Device	☐	Connect...

CANCEL

OK

Compartilhamento dos discos de ASM no nó 01, GNCASHTL08920:
 Consiste em alterar os discos da controller 1, Hard disk 3, 4 e 5 para:

Sharing: Multi-writer
 Disk Mode: Independent – Persistent

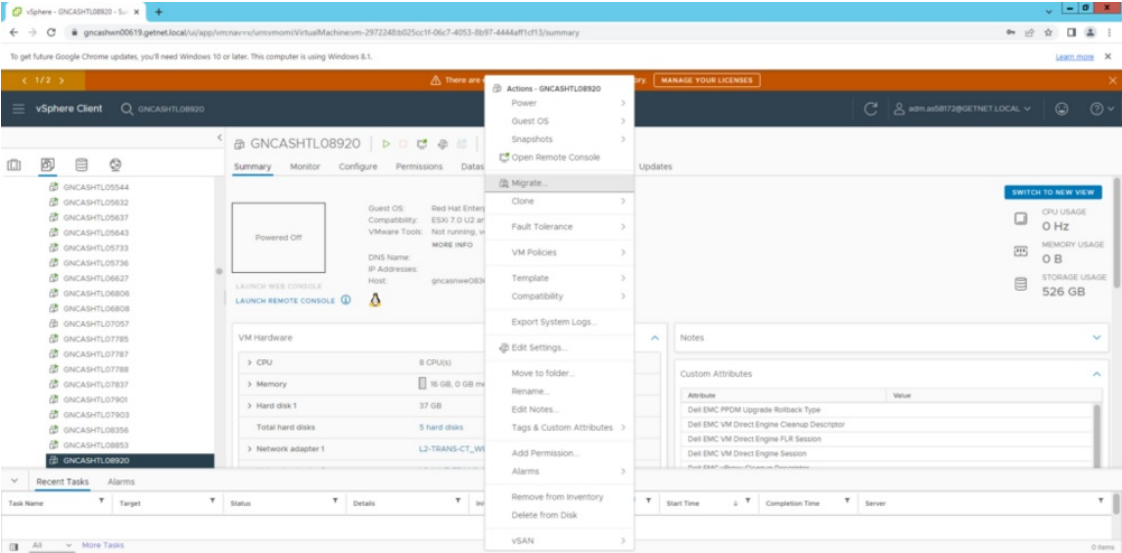
Edit Settings
GNCASHTL08920

> Hard disk 1	37 GB SCSI(0:0)	
> Hard disk 2	162 GB SCSI(0:1)	
> Hard disk 3 *	145	GB
Maximum Size	6.93 TB	
VM storage policy	Datastore Default	
Type	Thick Provision Eager Zeroed	
Sharing	Multi-writer	
Disk File	[DS_LINUX_SHOULD_PMX2] GNCASHTL08920/GNCASHTL08920_2.vmdk	
Shares	Normal	1000
Limit - IOPs	Unlimited	
Disk Mode	Independent - Persistent	
Virtual Device Node	SCSI controller 1	SCSI(1:0) Hard disk 3
> Hard disk 4	145 GB SCSI(1:1)	
> Hard disk 5	37 GB SCSI(1:2)	
> SCSI controller 0	VMware Paravirtual	

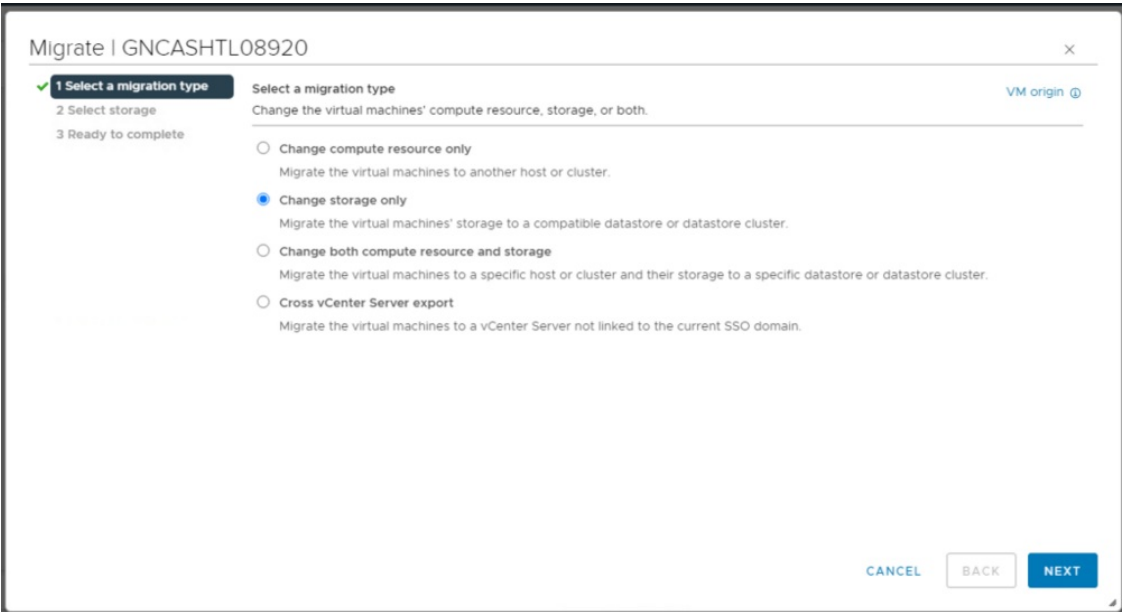
CANCEL

OK

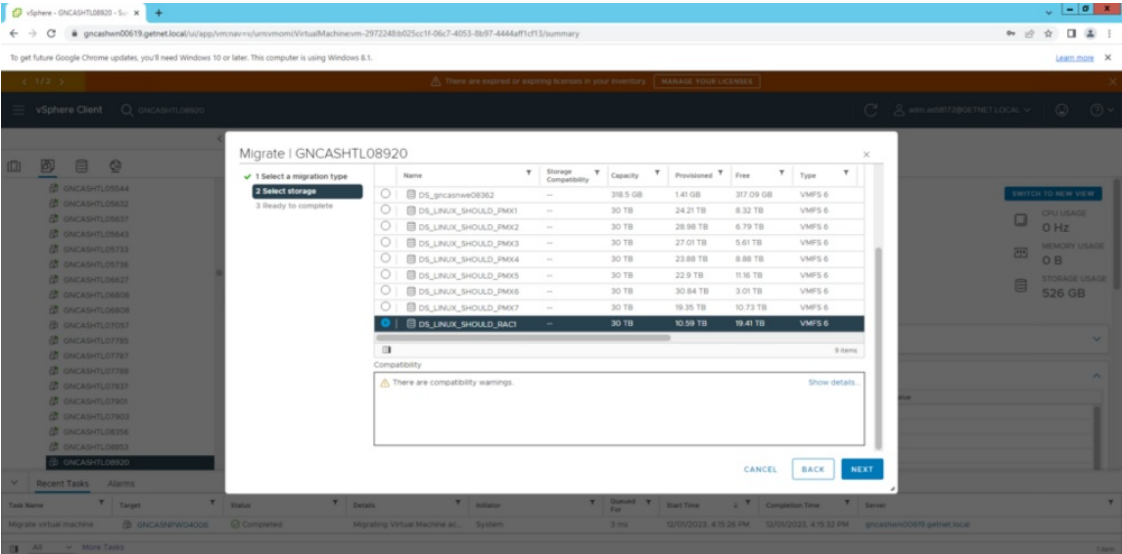
Necessário mover as VMS para o Cluster de RAC:
Vá em "Edit Settings" > Migrate..
NOTA: Fazer isso para as 3 VMS



Escolha a opção "Change Storage Only"



Escolha o Cluster de RAC:



Migrate | GNCASHTL08920

1 Select a migration type

2 Select storage

3 Ready to complete

Ready to complete

Verify that the information is correct and click Finish to start the migration.

VM origin

Migration Type	Change storage. Leave VM on the original compute resource
Virtual Machine	GNCASHTL08920
Storage	DS_LINUX_SHOULD_RAC1
Disk Format	Same format as source

CANCEL

BACK

FINISH

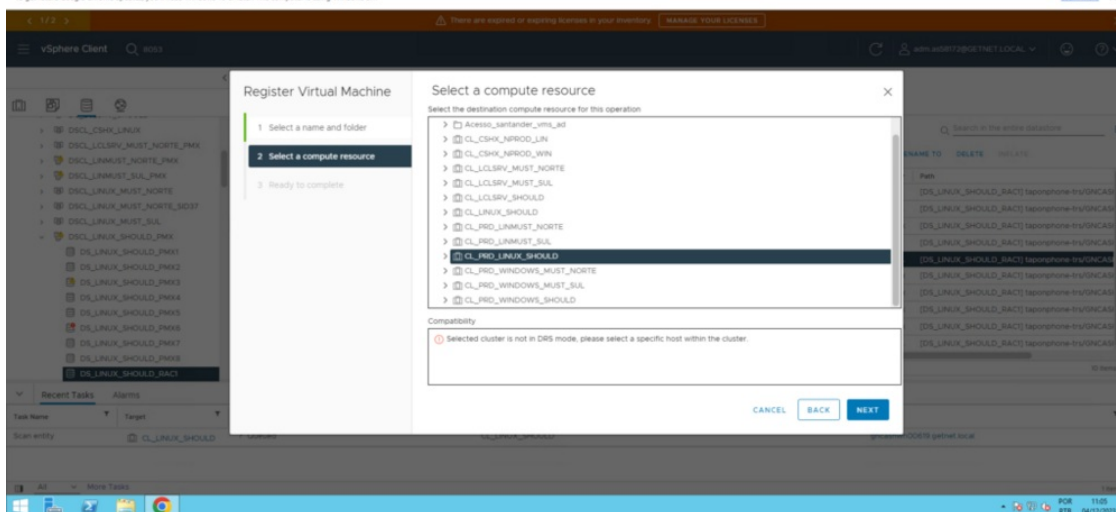
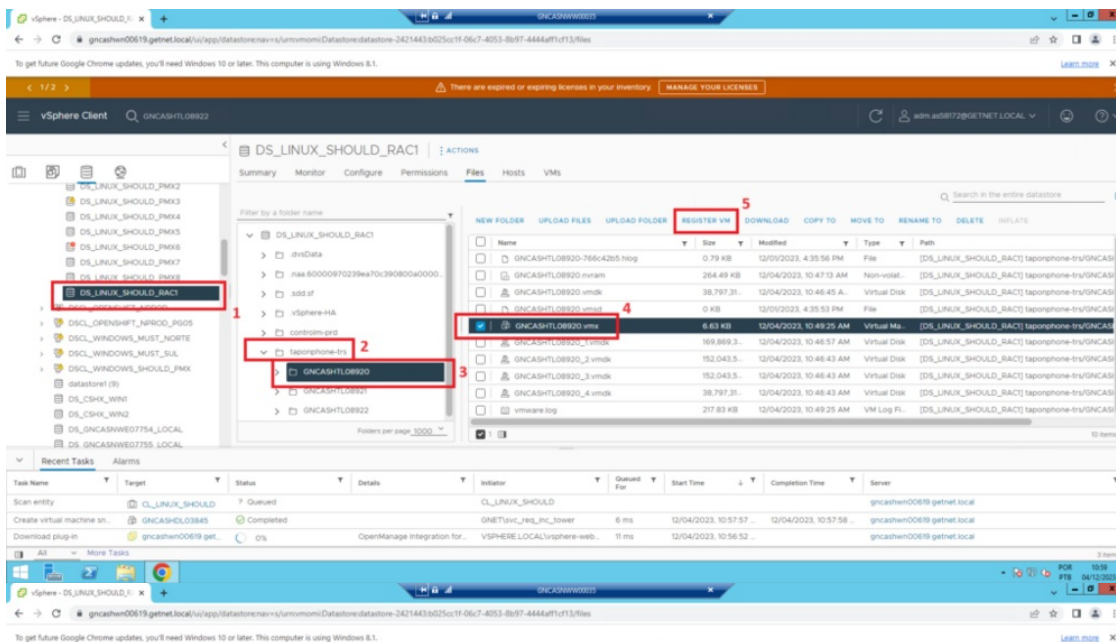
Um passo importante, para não haver problemas de apagarem as VMs, é criar a folder para os hosts:
Nome utilizado: tapophone-trs

Selecione as 3 VMs e clique em Move to:
Escolha a pasta criada.

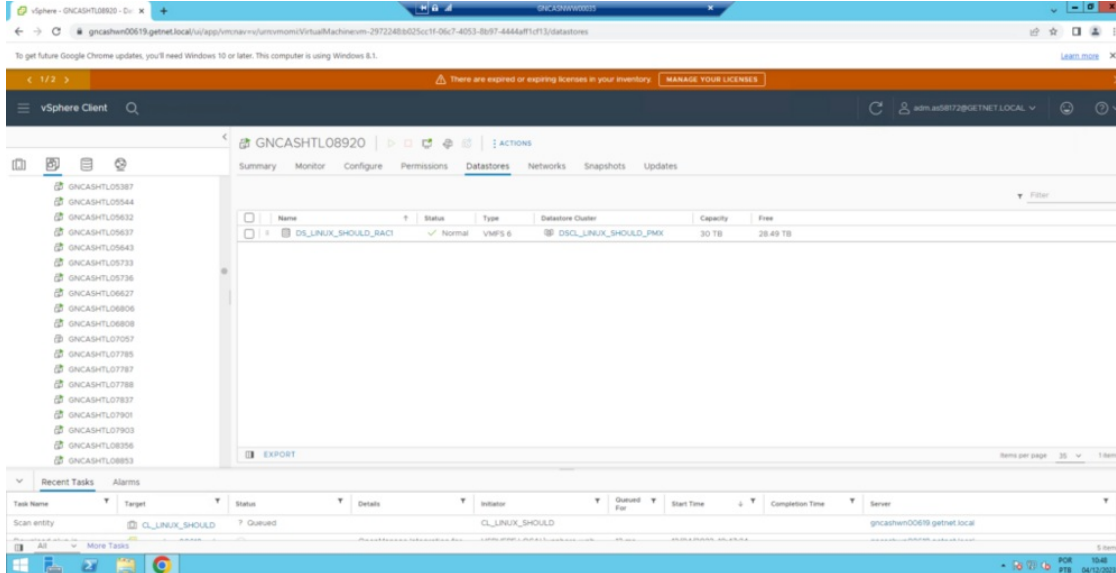
NOTA: ao mover as VMs para a pasta, costumou remove-las do inventory e readiciona-las de dentro do pasta do DataStore:

Core Assets/Documentation/GETNET -
DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

16

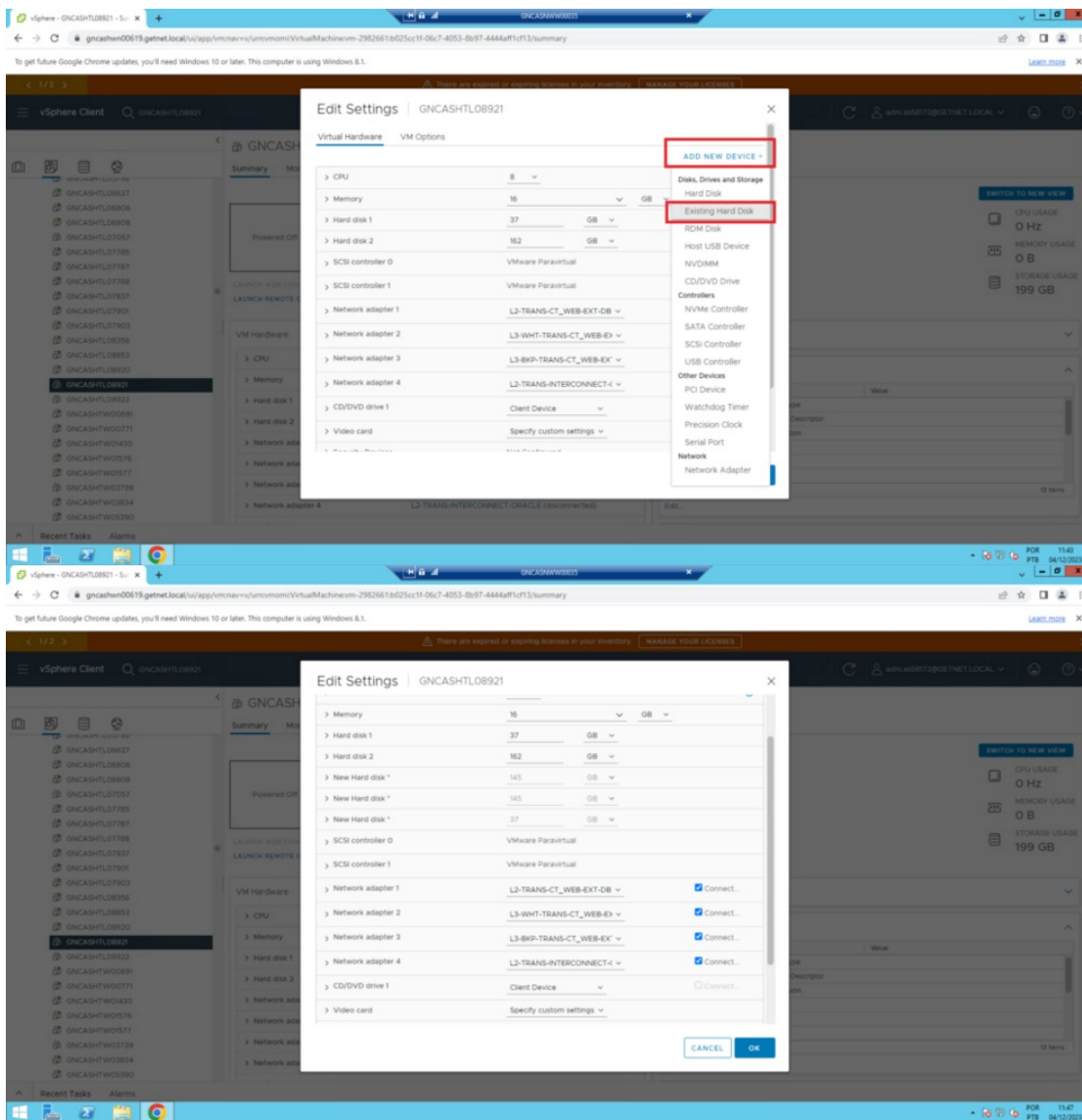


Após fazer o Start da VM, já é possível conferir que o host está no DS_LINUX_SHOULD_RAC1:



Compartilhando discos

Neste momento, deixe apenas a VM GNCASHTL08920 up, pois ela tem os discos. Faça adição dos discos nas VMs GNCASHTL08921 e GNCASHTL08922 com as VMs down.

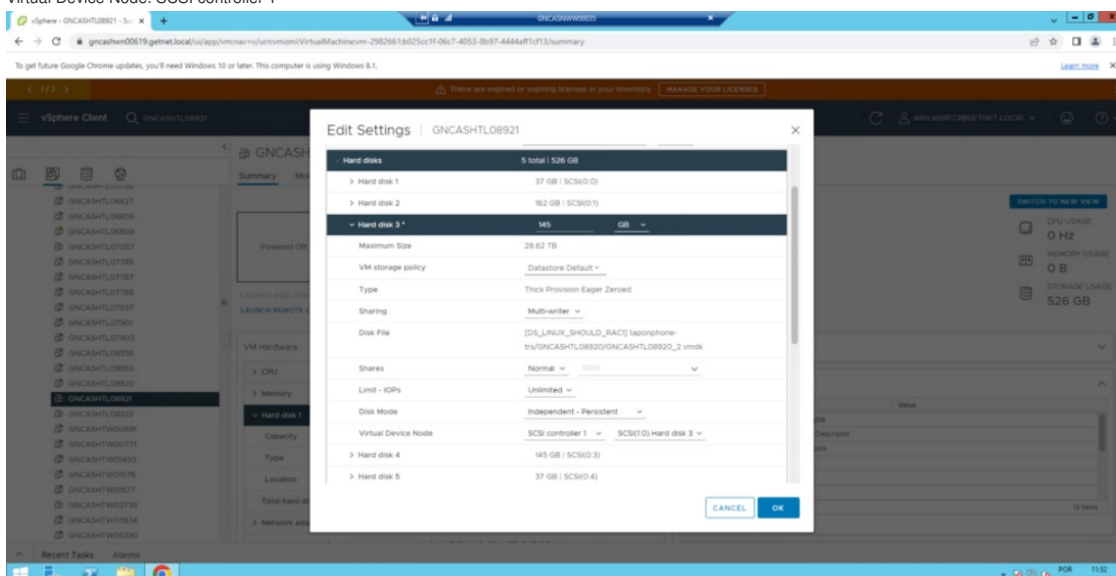


Com os discos adicionados, relembre de reforçar as configurações em cada disco nas duas VMs:

Sharing: Multi-writer

Disk Mode: Independent – Persistent

Virtual Device Node: SCSI controller 1



Só com todas as configurações feitas nos discos, podemos subir as VMs, interessante subir uma a uma para rastrear qual pode estar com erro de configuração.

NOTE que neste ponto, os discos aparecem nas 3 VMs:

Multi execution mode: commands are typed to all terminals (use Ctrl+Shift+Insert to paste)

Multi-paste

Exit multi-execution mode

Disk /dev/sdc: 145 GiB, 155682564480 bytes, 304087040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9a2f6561

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdc1		2048	76021759	76019712	36.36 GiB	83	Linux
/dev/sdc2		76021760	152043519	76021760	36.36 GiB	83	Linux
/dev/sdc3		152043520	228065279	76021760	36.36 GiB	83	Linux
/dev/sdc4		228065280	304087039	76021760	36.36 GiB	83	Linux

Disable this terminal from "GUI/CLI" mode

Disk /dev/sdd: 145 GiB, 155682564480 bytes, 304087040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xc1f740b2

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdd1		2048	76021759	76019712	36.36 GiB	83	Linux
/dev/sdd2		76021760	152043519	76021760	36.36 GiB	83	Linux
/dev/sdd3		152043520	228065279	76021760	36.36 GiB	83	Linux

Disable this terminal from "GUI/CLI" mode

Disk /dev/sde: 27 GiB, 28728447488 bytes, 77594016 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7b3ebaba

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sde1		2048	19398055	19396008	9.30 GiB	83	Linux
/dev/sde2		19398056	38797111	19399056	9.30 GiB	83	Linux
/dev/sde3		38797112	58195867	19398056	9.30 GiB	83	Linux
/dev/sde4		58195868	77594023	19398056	9.30 GiB	83	Linux

Disable this terminal from "GUI/CLI" mode

Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9a2f6561

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdc1		2048	76021759	76019712	36.36 GiB	83	Linux
/dev/sdc2		76021760	152043519	76021760	36.36 GiB	83	Linux
/dev/sdc3		152043520	228065279	76021760	36.36 GiB	83	Linux
/dev/sdc4		228065280	304087039	76021760	36.36 GiB	83	Linux

Disable this terminal from "GUI/CLI" mode

Disk /dev/sdd: 145 GiB, 155682564480 bytes, 304087040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xc1f740b2

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdd1		2048	76021759	76019712	36.36 GiB	83	Linux
/dev/sdd2		76021760	152043519	76021760	36.36 GiB	83	Linux
/dev/sdd3		152043520	228065279	76021760	36.36 GiB	83	Linux
/dev/sdd4		228065280	304087039	76021760	36.36 GiB	83	Linux

Disable this terminal from "GUI/CLI" mode

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobaxterm.net>

Limpeza dos discos apenas no nó 01:

Core Assets/Documentation/GETNET -
DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

19

```

[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdc1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.199072 s, 257 kB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdc2 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0132821 s, 3.9 MB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdc3 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0143289 s, 3.6 MB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdc4 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0178288 s, 2.9 MB/s
[root@GNCASHTL08920 ~]#
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdd1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.240812 s, 213 kB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdd2 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0249882 s, 2.0 MB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdd3 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0277899 s, 1.8 MB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sdd4 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0098233 s, 5.2 MB/s
[root@GNCASHTL08920 ~]#
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sde1 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.205266 s, 249 kB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sde2 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0291022 s, 1.8 MB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sde3 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0244742 s, 2.1 MB/s
[root@GNCASHTL08920 ~]# dd if=/dev/zero of=/dev/sde4 bs=512 count=100
100+0 records in
100+0 records out
51200 bytes (51 kB, 50 KiB) copied, 0.0144768 s, 3.5 MB/s

```

Criação dos discos de ASM:

```

[root@GNCASHTL08920 ~]# ./home/grid/grid_profile
[root@GNCASHTL08920 ~]# oracleasm createdisk ARCH01_TAPONPHONE /dev/sdc1
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk ARCH02_TAPONPHONE /dev/sdc2
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk ARCH03_TAPONPHONE /dev/sdc3
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk ARCH04_TAPONPHONE /dev/sdc4
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA01_AUX_TAPONPHONE /dev/sde1
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA02_AUX_TAPONPHONE /dev/sde2
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA03_AUX_TAPONPHONE /dev/sde3
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA04_AUX_TAPONPHONE /dev/sde4
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA01_TAPONPHONE /dev/sdd1
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA02_TAPONPHONE /dev/sdd2
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA03_TAPONPHONE /dev/sdd3
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]# oracleasm createdisk DATA04_TAPONPHONE /dev/sdd4
Writing disk header: done
Instantiating disk: done
[root@GNCASHTL08920 ~]#

[root@GNCASHTL08920 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...

[root@GNCASHTL08920 ~]# oracleasm listdisks
ARCH01_TAPONPHONE
ARCH02_TAPONPHONE
ARCH03_TAPONPHONE
ARCH04_TAPONPHONE
DATA01_AUX_TAPONPHONE
DATA01_TAPONPHONE
DATA02_AUX_TAPONPHONE
DATA02_TAPONPHONE
DATA03_AUX_TAPONPHONE
DATA03_TAPONPHONE
DATA04_AUX_TAPONPHONE
DATA04_TAPONPHONE
[root@GNCASHTL08920 ~]#

```

Veja que o ASM reconheceu todos os discos no nó 01, agora, no nó 02 e 03 é necessário apenas reconhecê-los. Com sucesso, todos encontrados:

```
[root@GNCASHTL08921 ~]# ./home/grid/grid_profile
[root@GNCASHTL08921 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
Instantiating disk "ARCH01_TAPONPHONE"
Instantiating disk "ARCH02_TAPONPHONE"
Instantiating disk "ARCH03_TAPONPHONE"
Instantiating disk "ARCH04_TAPONPHONE"
Instantiating disk "DATA01_TAPONPHONE"
Instantiating disk "DATA02_TAPONPHONE"
Instantiating disk "DATA03_TAPONPHONE"
Instantiating disk "DATA04_TAPONPHONE"
Instantiating disk "DATA01_AUX_TAPONPHONE"
Instantiating disk "DATA02_AUX_TAPONPHONE"
Instantiating disk "DATA03_AUX_TAPONPHONE"
Instantiating disk "DATA04_AUX_TAPONPHONE"
[root@GNCASHTL08921 ~]# oracleasm listdisks
ARCH01_TAPONPHONE
ARCH02_TAPONPHONE
ARCH03_TAPONPHONE
ARCH04_TAPONPHONE
DATA01_AUX_TAPONPHONE
DATA01_TAPONPHONE
DATA02_AUX_TAPONPHONE
DATA02_TAPONPHONE
DATA03_AUX_TAPONPHONE
DATA03_TAPONPHONE
DATA04_AUX_TAPONPHONE
DATA04_TAPONPHONE
```

```
[root@GNCASHTL08922 ~]# ./home/grid/grid_profile
[root@GNCASHTL08922 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
Instantiating disk "ARCH01_TAPONPHONE"
Instantiating disk "ARCH02_TAPONPHONE"
Instantiating disk "ARCH03_TAPONPHONE"
Instantiating disk "ARCH04_TAPONPHONE"
Instantiating disk "DATA01_AUX_TAPONPHONE"
Instantiating disk "DATA02_AUX_TAPONPHONE"
Instantiating disk "DATA03_AUX_TAPONPHONE"
Instantiating disk "DATA04_AUX_TAPONPHONE"
Instantiating disk "DATA01_TAPONPHONE"
Instantiating disk "DATA02_TAPONPHONE"
Instantiating disk "DATA03_TAPONPHONE"
Instantiating disk "DATA04_TAPONPHONE"
[root@GNCASHTL08922 ~]# oracleasm listdisks
ARCH01_TAPONPHONE
ARCH02_TAPONPHONE
ARCH03_TAPONPHONE
ARCH04_TAPONPHONE
DATA01_AUX_TAPONPHONE
DATA01_TAPONPHONE
DATA02_AUX_TAPONPHONE
DATA02_TAPONPHONE
DATA03_AUX_TAPONPHONE
DATA03_TAPONPHONE
DATA04_AUX_TAPONPHONE
DATA04_TAPONPHONE
[root@GNCASHTL08922 ~]#
```

Copia dos binários

NOTA: os arquivos dos binários do Grid e do Oracle só são necessários no nó 01, porém, baixamos todos nos dois nós.

```
[grid@GNCASHTL08920 ~]$ . grid_profile
[grid@GNCASHTL08920 ~]$ cd /u02/oracle/patches/
```



```

[grid@GNCASHTL08920 ~]$ cd /u02/oracle/patches/
[grid@GNCASHTL08920 patches]$ wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_grid_home.zip
--2023-12-04 12:20:20-- https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_grid_home.zip
Resolving repository.getnet.com.br (repository.getnet.com.br)... 22.248.7.106
Connecting to repository.getnet.com.br (repository.getnet.com.br)|22.248.7.106|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2889184573 (2.7G) [application/zip]
Saving to: 'LINUX.X64_193000_grid_home.zip'

LINUX.X64_193000_grid_home.z 100%[=====>] 2.69G 21.3MB/s in 2m 23s

2023-12-04 12:22:44 (19.2 MB/s) - 'LINUX.X64_193000_grid_home.zip' saved [2889184573/2889184573]

[grid@GNCASHTL08920 patches]$ wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_db_home.zip
--2023-12-04 12:23:25-- https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/LINUX.X64_193000_db_home.zip
Resolving repository.getnet.com.br (repository.getnet.com.br)... 22.248.7.106
Connecting to repository.getnet.com.br (repository.getnet.com.br)|22.248.7.106|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3059705302 (2.8G) [application/zip]
Saving to: 'LINUX.X64_193000_db_home.zip'

LINUX.X64_193000_db_home.zip 100%[=====>] 2.85G 30.6MB/s in 94s

2023-12-04 12:24:59 (30.9 MB/s) - 'LINUX.X64_193000_db_home.zip' saved [3059705302/3059705302]

[grid@GNCASHTL08920 patches]$ wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
--2023-12-04 12:26:05-- https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p6880880_190000_Linux-x86-64.zip
Resolving repository.getnet.com.br (repository.getnet.com.br)... 22.248.7.106
Connecting to repository.getnet.com.br (repository.getnet.com.br)|22.248.7.106|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 127774864 (122M) [application/zip]
Saving to: 'p6880880_190000_Linux-x86-64.zip'

p6880880_190000_Linux-x86-64 100%[=====>] 121.86M 43.0MB/s in 2.8s

2023-12-04 12:26:08 (43.0 MB/s) - 'p6880880_190000_Linux-x86-64.zip' saved [127774864/127774864]

[grid@GNCASHTL08920 patches]$ wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35642822_190000_Linux-x86-64.zip
--2023-12-04 12:26:28-- https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35642822_190000_Linux-x86-64.zip
Resolving repository.getnet.com.br (repository.getnet.com.br)... 22.248.7.106
Connecting to repository.getnet.com.br (repository.getnet.com.br)|22.248.7.106|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3084439097 (2.9G) [application/zip]
Saving to: 'p35642822_190000_Linux-x86-64.zip'

p35642822_190000_Linux-x86-6 100%[=====>] 2.87G 76.4MB/s in 50s

2023-12-04 12:27:18 (59.2 MB/s) - 'p35642822_190000_Linux-x86-64.zip' saved [3084439097/3084439097]

[grid@GNCASHTL08920 patches]$ wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35648110_190000_Linux-x86-64.zip
--2023-12-04 12:30:59-- https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35648110_190000_Linux-x86-64.zip
Resolving repository.getnet.com.br (repository.getnet.com.br)... 22.248.7.106
Connecting to repository.getnet.com.br (repository.getnet.com.br)|22.248.7.106|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 127350205 (121M) [application/zip]
Saving to: 'p35648110_190000_Linux-x86-64.zip'

p35648110_190000_Linux-x86-6 100%[=====>] 121.45M 41.6MB/s in 2.9s

2023-12-04 12:31:02 (41.6 MB/s) - 'p35648110_190000_Linux-x86-64.zip' saved [127350205/127350205]

[grid@GNCASHTL08920 patches]$ wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35638318_190000_Linux-x86-64.zip
--2023-12-04 12:31:30-- https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p35638318_190000_Linux-x86-64.zip
Resolving repository.getnet.com.br (repository.getnet.com.br)... 22.248.7.106
Connecting to repository.getnet.com.br (repository.getnet.com.br)|22.248.7.106|:443... connected.

```

```
Connecting to repository.getnet.com.br (repository.getnet.com.br)[22.246.7.100]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 121857143 (116M) [application/zip]
Saving to: 'p35638318_190000_Linux-x86-64.zip'

p35638318_190000_Linux-x86-6 100%[=====>] 116.21M  37.2MB/s   in 3.2s

2023-12-04 12:31:33 (36.2 MB/s) - 'p35638318_190000_Linux-x86-64.zip' saved [121857143/121857143]

[grid@GNCASHTL08920 patches]$
```

Ajustes dos Profiles

Profiles do Grid:

```
[grid@GNCASHTL08920 ~]$ cat grid_profile
export ORACLE_BASE=/u01/app/grid
export ORACLE_HOME=/u01/app/19.21.0/grid
export ORACLE_SID=+ASM1
export PATH=/u01/app/19.21.0/grid/bin:$PATH
```

```
[grid@GNCASHTL08921 ~]$ cat grid_profile
export ORACLE_BASE=/u01/app/grid
export ORACLE_HOME=/u01/app/19.21.0/grid
export ORACLE_SID=+ASM2
export PATH=/u01/app/19.21.0/grid/bin:$PATH
```

```
[grid@GNCASHTL08922 ~]$ cat grid_profile
export ORACLE_BASE=/u01/app/grid
export ORACLE_HOME=/u01/app/19.21.0/grid
export ORACLE_SID=+ASM3
```

Profile do Oracle:

```
[oracle@GNCASHTL08920 ~]$ cat db_profile
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/19.21.0/dbhome_1
export ORACLE_SID=TAPONPHONE1
export PATH=/u01/app/oracle/product/19.21.0/dbhome_1/bin:$PATH
export NLS_LANG="AMERICAN_AMERICA.WE8MSWIN1252"
```

```
[oracle@GNCASHTL08921 ~]$ cat db_profile
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/19.21.0/dbhome_1
export ORACLE_SID=TAPONPHONE2
export PATH=/u01/app/oracle/product/19.21.0/dbhome_1/bin:$PATH
export NLS_LANG="AMERICAN_AMERICA.WE8MSWIN1252"
```

```
[oracle@GNCASHTL08922 ~]$ cat db_profile
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/19.21.0/dbhome_1
export ORACLE_SID=TAPONPHONE3
export PATH=/u01/app/oracle/product/19.21.0/dbhome_1/bin:$PATH
export NLS_LANG="AMERICAN_AMERICA.WE8MSWIN1252"
```

Criação dos diretórios nas 3 máquinas


```
mkdir -p /u01/app/oracle/product/19.21.0/dbhome_1
mkdir -p /u01/app/19.21.0/grid
chown -R grid:oinstall /u01/app/19.21.0/
chown -R oracle:oinstall /u01/app/oracle/product/19.21.0
mkdir -p /u01/app/grid_19c
mkdir -p /u01/app/oracle_19c
chown -R grid:oinstall /u01/app/grid/
chown -R oracle:oinstall /u01/app/oracle/
chmod 775 /u01/app/oracle_19c/
chmod 775 /u01/app/grid_19c/
mkdir -p /u01/app/orainventory
chown grid:oinstall /u01/app/orainventory
chmod 775 /u01/app/orainventory
mkdir -p /u01/app/grid/
chown -R grid:oinstall /u01/app/grid/
```

Unzip dos binários

Unzip dos Binários do Grid Infrastructure no nó 01

```
[grid@GNCASHTL08920 ~]$ . grid_profile
[grid@GNCASHTL08920 ~]$ cd $ORACLE_HOME
[grid@GNCASHTL08920 grid]$ pwd
/u01/app/19.21.0/grid
[grid@GNCASHTL08920 grid]$ cp /u02/oracle/patches/LINUX.X64_193000_grid_home.zip .
[grid@GNCASHTL08920 grid]$ unzip LINUX.X64_193000_grid_home.zip
...
```

Ajustes do OPatch na home do Grid Infrastructure

```
[grid@GNCASHTL08920 grid]$ rm -rf OPatch/
[grid@GNCASHTL08920 grid]$ cp /u02/oracle/patches/p6880880_190000_Linux-x86-64.zip .
[grid@GNCASHTL08920 grid]$ unzip p6880880_190000_Linux-x86-64.zip
...
```

Unzip dos binários do Oracle Database

```
[oracle@GNCASHTL08920 ~]$ . db_profile
[oracle@GNCASHTL08920 ~]$ cd $ORACLE_HOME
[oracle@GNCASHTL08920 dbhome_1]$ pwd
/u01/app/oracle/product/19.21.0/dbhome_1
[oracle@GNCASHTL08920 dbhome_1]$ cp /u02/oracle/patches/LINUX.X64_193000_db_home.zip .
[oracle@GNCASHTL08920 dbhome_1]$ unzip LINUX.X64_193000_db_home.zip
...
```

Ajustes do Opatch na home do Oracle Database

```
[oracle@GNCASHTL08920 dbhome_1]$ rm -rf OPatch/
[oracle@GNCASHTL08920 dbhome_1]$ cp /u02/oracle/patches/p6880880_190000_Linux-x86-64.zip .
[oracle@GNCASHTL08920 dbhome_1]$ unzip p6880880_190000_Linux-x86-64.zip
...
[oracle@GNCASHTL08920 dbhome_1]$ rm -rf *.zip
```

Criação das Chaves entre Hosts, user Oracle e Grid

Esse é um ponto muito chato na configuração dos RACs, é muito bugado a criação de chaves feita pelo Oracle, sendo assim, fazemos nos 2 nós e duplicamos as linhas.

Incluir o grupo oinstall no ssh (/etc/ssh/sshd_config):

```
vi /etc/ssh/sshd_config

# Allowed Groups
AllowGroups oinstall GL_Acesso_SSH GG_Acesso_Infra-Estrutura_TS GG_GNCASHTL08922_USR GG_GNCASHTL08922_ADM
```

Importante colocar as senhas do Oracle e Grid nos 3 nós iguais:

```
[root@GNCASHTL08921 ~]# passwd grid
Changing password for user grid.
New password:
BAD PASSWORD: The password contains less than 4 character classes
Retype new password:
passwd: all authentication tokens updated successfully.
```

```
[root@GNCASHTL08921 ~]# passwd oracle
Changing password for user oracle.
New password:
BAD PASSWORD: The password contains less than 4 character classes
Retype new password:
passwd: all authentication tokens updated successfully.
```

Configurando as chaves entre os hosts:

NOTA: jeito rápido de configurar as chaves de ssh para grid e oracle:

Nó principal:

```
[grid@GNCASHTL08920 scripts]$ pwd
/u01/app/19.21.0/grid/oui/prov/resources/scripts
```

Nós secundários:

```
[grid@GNCASHTL08922 scripts]$ pwd
/u02/oracle/patches/oui/prov/resources/scripts
```

```
[grid@GNCASHTL08921 scripts]$ pwd
/u02/oracle/patches/oui/prov/resources/scripts
```

Rode o script:

```
./sshUserSetup.sh -user grid -hosts "GNCASHTL08920 GNCASHTL08921 GNCASHTL08922" -noPromptPassphrase
```

Passe a senha configurada para o user grid.

Com o comando cima, ca host, vai gerar uma chave de ssh, exemplo:

```
[grid@GNCASHTL08921 scripts]$ vi /home/grid/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDmDMD/R9LPE1icPsJz1F6NMOI8Fc4e38F2TvSnXYLBlvm0z5ahOEvhgXDCoAR4MrkexdrEIDgl
0Q0tFVQWTlwbT0mVosl2Clcf79nbtlaFYMrvZ9sRUjprhBxIwYw64e3OW13HiTC5jUnUdudijszlDIYCsBztYkNZ1I7a4Lgw4dw== grid@GNCASHTL
08921
```

Copie para os 3 servidores, as linhas geradas:

```
[grid@GNCASHTL08921 scripts]$ vi /home/grid/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDmDMD/R9LPE1icPsJz1F6NMOI8Fc4e38F2TvSnXYLBlvm0z5ahOEvhgXDCoAR4MrkexdrEIDgl
0Q0tFVQWTlwbT0mVosl2Clcf79nbtlaFYMrvZ9sRUjprhBxIwYw64e3OW13HiTC5jUnUdudijszlDIYCsBztYkNZ1I7a4Lgw4dw== grid@GNCASHTL0
8920
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDmDMD/R9LPE1icPsJz1F6NMOI8Fc4e38F2TvSnXYLBlvm0z5ahOEvhgXDCoAR4MrkexdrEIDgl
mXaa5i/s60bkcb3+2MxdDxTF5aBVBiiPocJY5b9Zp5E5z5jo4dO7BbU7X/7zlr7Jlm73IAOpEoM41ZDrcso6El/tyF4qCwk6DGvQ== grid@GNCASHTL
08921
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDmDMD/R9LPE1icPsJz1F6NMOI8Fc4e38F2TvSnXYLBlvm0z5ahOEvhgXDCoAR4MrkexdrEIDgl
sln2k+8pmvoqWs2PqcsulzchFkiYqUBSf3InY+BKrx1pD+PcnTa+4DgJzF5UV5hS5YD7ORDVJEnVnHUMHDBF/9qkPXeUMBEXQ== grid@GNCA
SHTL08922
```

Depois disso, teste o comando date, ele deve retornar a data sem pedir senha:

```
[grid@GNCASHTL08921 scripts]$ /usr/bin/ssh -x -l grid GNCASHTL08920 date
Tue Dec  5 17:02:29 -03 2023
[grid@GNCASHTL08921 scripts]$ /usr/bin/ssh -x -l grid GNCASHTL08921 date
Tue Dec  5 17:02:41 -03 2023
[grid@GNCASHTL08921 scripts]$ /usr/bin/ssh -x -l grid GNCASHTL08922 date
Tue Dec  5 17:02:45 -03 2023
```

A mesma configuração se dá para o user oracle

Nó principal:

```
[oracle@GNCASHTL08920 scripts]$ pwd
/u01/app/19.21.0/grid/oui/prov/resources/scripts
```

Nós secundários:

```
[oracle@GNCASHTL08921 scripts]$ pwd
/u02/oracle/patches/oui/prov/resources/scripts
```

```
[oracle@GNCASHTL08922 scripts]$ pwd
/u02/oracle/patches/oui/prov/resources/scripts
```

Rode o script:

```
./sshUserSetup.sh -user oracle -hosts "GNCASHTL08920 GNCASHTL08921 GNCASHTL08922" -noPromptPassphrase
```

Passa a senha configurada para o user oracle.

Com o comando cima, ca host, vai gerar uma chave de ssh, exemplo:

```
[oracle@GNCASHTL08920 scripts]$ cat /home/oracle/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDQ9nf4IDmj1Gfxhu7d9nNjK6lCs5Wg0atkJb1hjePp5CRq/nVvKmFpl0EJ6xP/sGGjR8C0NLs1nLHzivBJ2iJ493xCNtCR/7krF8HZF04IPqrTaztfY0YmlzzjG6UlcXbRqithqSydTAJYH/rUhf4LzO0LAVdQBw1saoDjtcQ== oracle@GNCASHTL08920
```

Copie para os 3 servidores, as linhas geradas:

```
[oracle@GNCASHTL08920 scripts]$ cat /home/oracle/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDQ9nf4IDmj1Gfxhu7d9nNjK6lCs5Wg0atkJb1hjePp5CRq/nVvKmFpl0EJ6xP/sGGjR8C0NLs1nLHzivBJ2iJ493xCNtCR/7krF8HZF04IPqrTaztfY0YmlzzjG6UlcXbRqithqSydTAJYH/rUhf4LzO0LAVdQBw1saoDjtcQ== oracle@GNCASHTL08920
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQCq/YppAE1AwJRu808jWYFqGYlvg09Oloj1WFxHmZtg5irZD6mCS0GUi4jNXBRRL4BGn+8aa64OoQl8dM3HfBwX4W5Qsfy8Pdg47NdayWX0YSm8Dafwkp9yk9N0KNRshEeg3gMbB9D9Wlti1FW9YgAoMX+FKEXA3WskFCBTGmsqQ== oracle@GNCASHTL08921
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDAgEOMQ/Hf/OdtRly7s3+uQESZbr5RpJ9xWj43Nb0a6TLEIUq5uYjwIRI6FcgtlwmJFn6gnZR0kSAkF7H0J9i/udmzvneRdgNfTLESkRMMVj5Z3gvRH6ONf1843LbNwBri2LbssGyVqLz2rxCNyGzPnKSUYlbgPZkz/tj9hj8w== oracle@GNCASHTL08922
```

Depois disso, teste o comando date, ele deve retornar a data sem pedir senha:

```
[oracle@GNCASHTL08920 scripts]$ /usr/bin/ssh -x -l oracle GNCASHTL08920 date
Tue Dec  5 17:12:15 -03 2023
[oracle@GNCASHTL08920 scripts]$ /usr/bin/ssh -x -l oracle GNCASHTL08921 date
Tue Dec  5 17:12:19 -03 2023
[oracle@GNCASHTL08920 scripts]$ /usr/bin/ssh -x -l oracle GNCASHTL08922 date
Tue Dec  5 17:12:24 -03 2023
```

Demais ajustes de S.O.

Ajustes do umask:

Adicionar a linha "umask 022" nos users oracle e grid nos 3 nós:

```
[oracle@GNCASHTL08921 ~]$ pwd
/home/oracle
[oracle@GNCASHTL08921 ~]$ vi ~/.bashrc
```

O mesmo para o user grid:

```
[grid@GNCASHTL08921 ~]$ pwd
/home/grid
[grid@GNCASHTL08921 ~]$ vi ~/.bashrc
```

Ajustar o /etc/resolv.conf:

```
[grid@GNCASHTL08921 ~]$ cat /etc/resolv.conf
domain getnet.local
nameserver 180.104.1.12
nameserver 180.104.7.15
nameserver 180.104.7.16
options timeout:1
```

Install do Grid Infrastructure

Export do display, apenas pelo cyberark:

```
[adm.as58172@GNCASHTL08920 ~]$ xauth list
GNCASHTL08920/unix:10 MIT-MAGIC-COOKIE-1 44e7231cf05ccf49c151fd1742aa0f61
GNCASHTL08920/unix:11 MIT-MAGIC-COOKIE-1 9a0178333db9bcda9f4c214cde7e00d3
[adm.as58172@GNCASHTL08920 ~]$ sudo su -
[sudo] password for adm.as58172:
Last login: Tue Dec 5 16:51:25 -03 2023
[root@GNCASHTL08920 ~]# su - grid
Last login: Tue Dec 5 17:14:50 -03 2023 on pts/0
[grid@GNCASHTL08920 ~]$ . grid_profile
[grid@GNCASHTL08920 ~]$ xauth add GNCASHTL08920/unix:11 MIT-MAGIC-COOKIE-1 9a0178333db9bcda9f4c214cde7e00d3
xauth: file /home/grid/.Xauthority does not exist
[grid@GNCASHTL08920 ~]$ xauth add GNCASHTL08920/unix:11 MIT-MAGIC-COOKIE-1 9a0178333db9bcda9f4c214cde7e00d3
[grid@GNCASHTL08920 ~]$ export DISPLAY=localhost:11.0
```

Export de uma variável que simula que o S.O. não é tão novo:

```
[grid@GNCASHTL08920 ~]$ export CV_ASSUME_DISTID=OEL8.1
```

Neste ponto, estou adicionando um patch nos instaladores, antes chamar o instalador em si pois descobrimos um bug:
INS-06006 GI RunInstaller Fails If OpenSSH Is Upgraded to 8.x (Doc ID 2555697.1)

```
[grid@GNCASHTL08920 ~]$ cd /u02/oracle/patches/
[grid@GNCASHTL08920 patches]$ wget https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p30159782_195000OCWRU_L
linux-x86-64.zip
--2023-12-05 17:22:22-- https://repository.getnet.com.br/binarios/oracle/odb/PST19C_TEMPLATE/p30159782_195000OCWRU_Linux-x86-64.zip
Resolving repository.getnet.com.br (repository.getnet.com.br)... 22.248.7.106
Connecting to repository.getnet.com.br (repository.getnet.com.br)|22.248.7.106|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 692329271 (660M) [application/zip]
Saving to: 'p30159782_195000OCWRU_Linux-x86-64.zip'

p30159782_195000OCWRU_Linux-x86-64.zip      100%[=====] 660.26M  21.0MB/s   in 35s

2023-12-05 17:22:57 (18.8 MB/s) - 'p30159782_195000OCWRU_Linux-x86-64.zip' saved [692329271/692329271]

[grid@GNCASHTL08920 patches]$ unzip p30159782_195000OCWRU_Linux-x86-64.zip
...

[grid@GNCASHTL08920 patches]$ cd $ORACLE_HOME
[grid@GNCASHTL08920 grid]$ ./gridSetup.sh -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /tmp/GridSetupActions2023-12-05_05-24-06PM/installerPatchActions_2023-12-05_05-24-06PM.log
Launching Oracle Grid Infrastructure Setup Wizard...

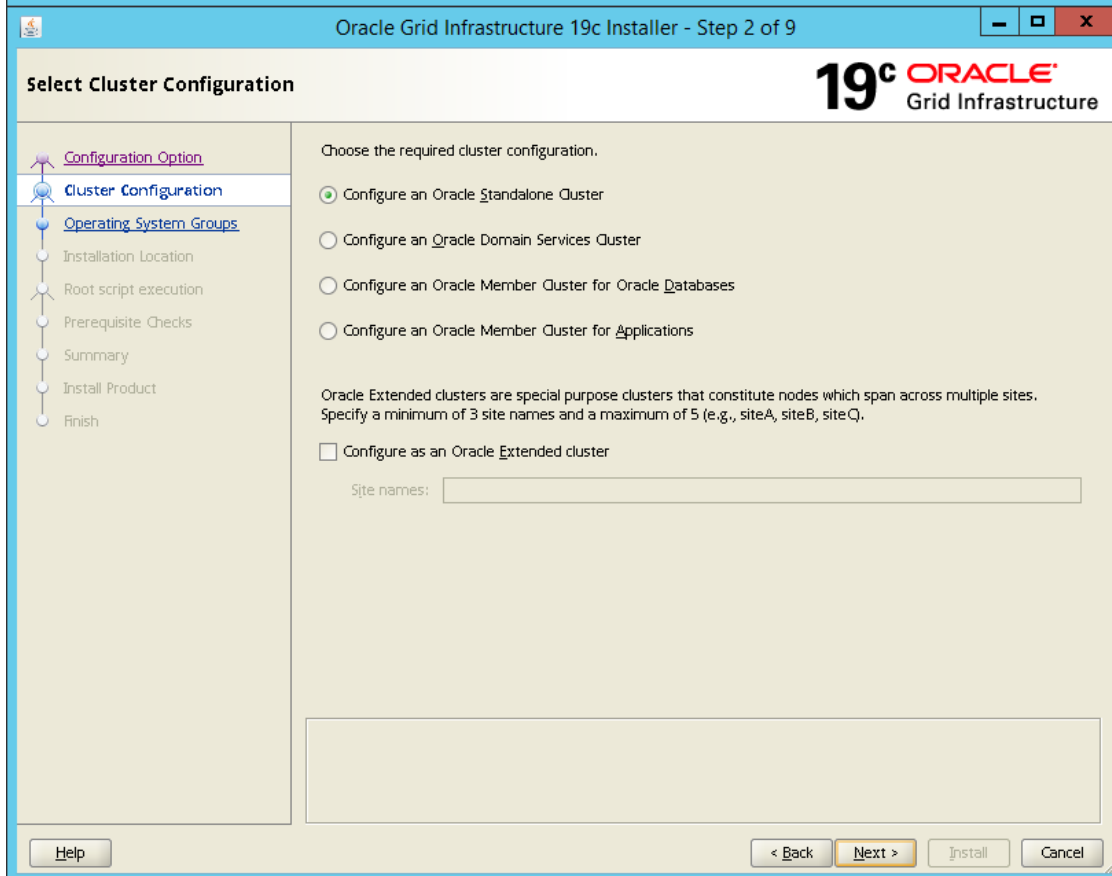
[FATAL] [INS-40426] Grid installation option has not been specified.
ACTION: Specify the valid installation option.
```

Agora sim, o Grid Setup:

```
[grid@GNCASHTL08920 grid]$ ./gridSetup.sh
ERROR: Unable to verify the graphical display setup. This application requires X display. Make sure that xdpinfo exist under PATH variable.
Launching Oracle Grid Infrastructure Setup Wizard...

The response file for this session can be found at:
/u01/app/19.21.0/grid/install/response/grid_2023-12-05_06-00-30PM.rsp

You can find the log of this install session at:
/tmp/GridSetupActions2023-12-05_06-00-30PM/gridSetupActions2023-12-05_06-00-30PM.log
Moved the install session logs to:
/u01/app/orainventory/logs/GridSetupActions2023-12-05_06-00-30PM
```



Oracle Grid Infrastructure 19c Installer - Step 3 of 17

Grid Plug and Play Information

Configuration Option

Cluster Configuration

Grid Plug and Play

Cluster Node Information

Network Interface Usage

Storage Option

Create Grid Infrastructure Management Repository

Grid Infrastructure Management Repository

Create ASM Disk Group

ASM Password

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Single Client Access Name (SCAN) allows clients to use one name in connection strings to connect to the cluster as a whole. Client connect requests to the SCAN name can be handled by any cluster node.

☒ Create Local SCAN

Cluster Name:

SCAN Name:

SCAN Port:

☐ Use Shared SCAN

SCAN Client Data:

☐ Configure GNS

☐ Configure nodes Virtual IPs as assigned by the Dynamic Networks

☒ Create a new GNS

GNS VIP Address:

GNS Sub Domain:

☐ Use Shared GNS

GNS Client Data:

Help

< Back

Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 4 of 17

Cluster Node Information

Configuration Option

Cluster Configuration

Grid Plug and Play

Cluster Node Information

Network Interface Usage

Storage Option

Create Grid Infrastructure Management Repository

Grid Infrastructure Management Repository

Create ASM Disk Group

ASM Password

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Provide the list of nodes to be managed by Oracle Grid Infrastructure with their Public Hostname and Virtual Hostname.

Public Hostname	Virtual Hostname
GNCASHTL08920	GNCASHTL08920-vip
GNCASHTL08921	GNCASHTL08921-vip
GNCASHTL08922	GNCASHTL08922-vip

SSH connectivity...

Use Cluster Configuration File...

Add...

Edit...

Remove

Help

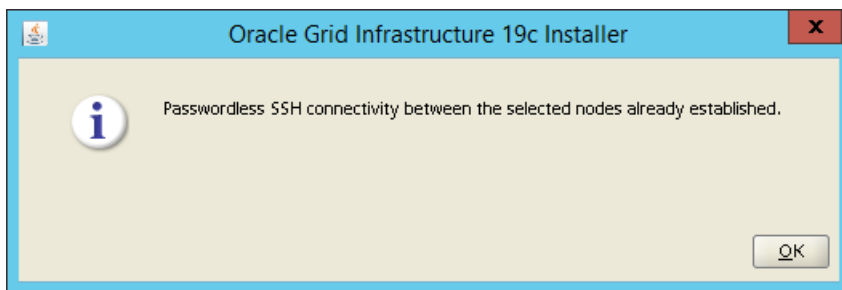
< Back

Next >

Install

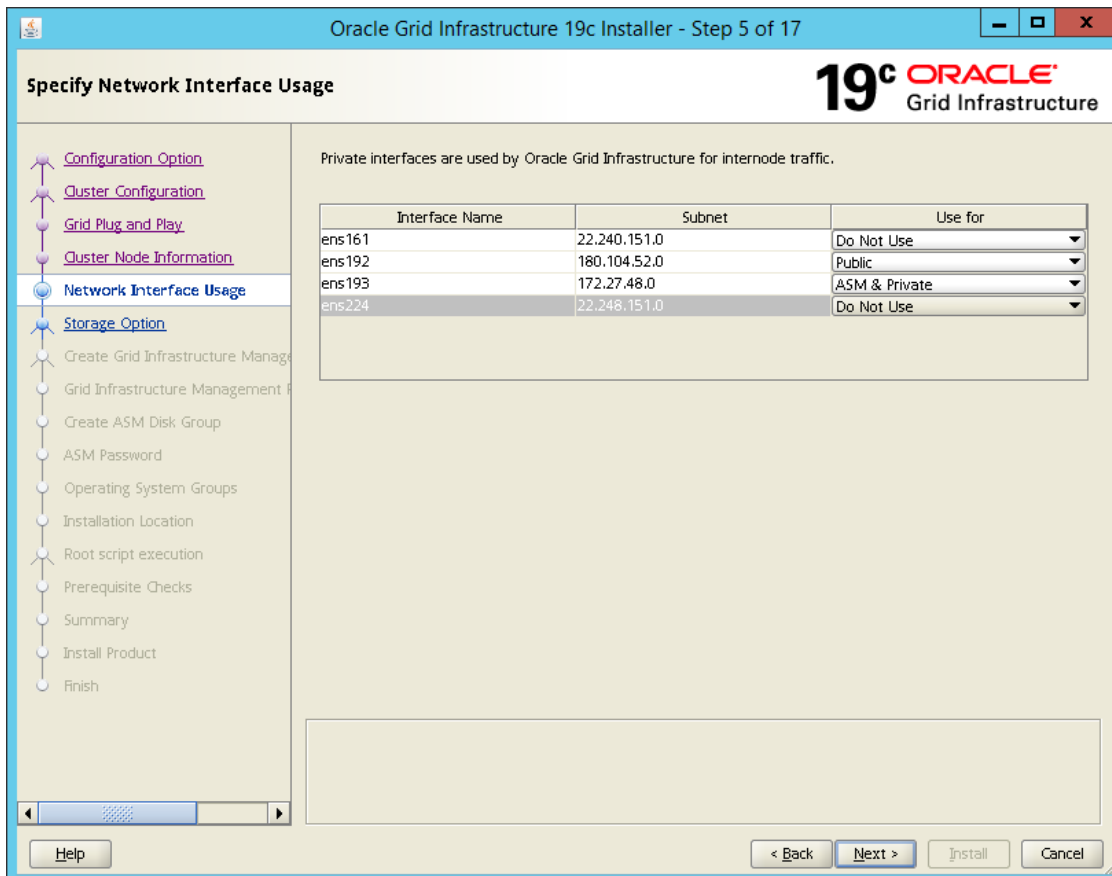
Cancel

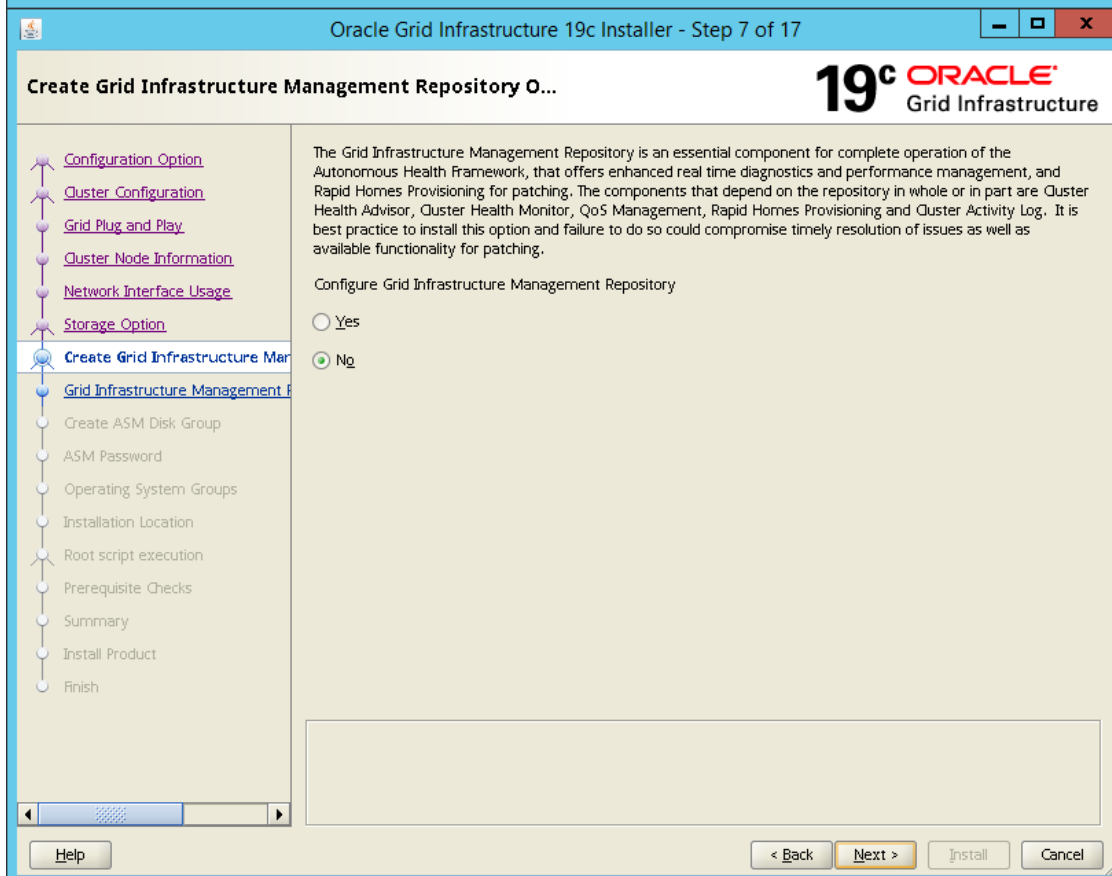
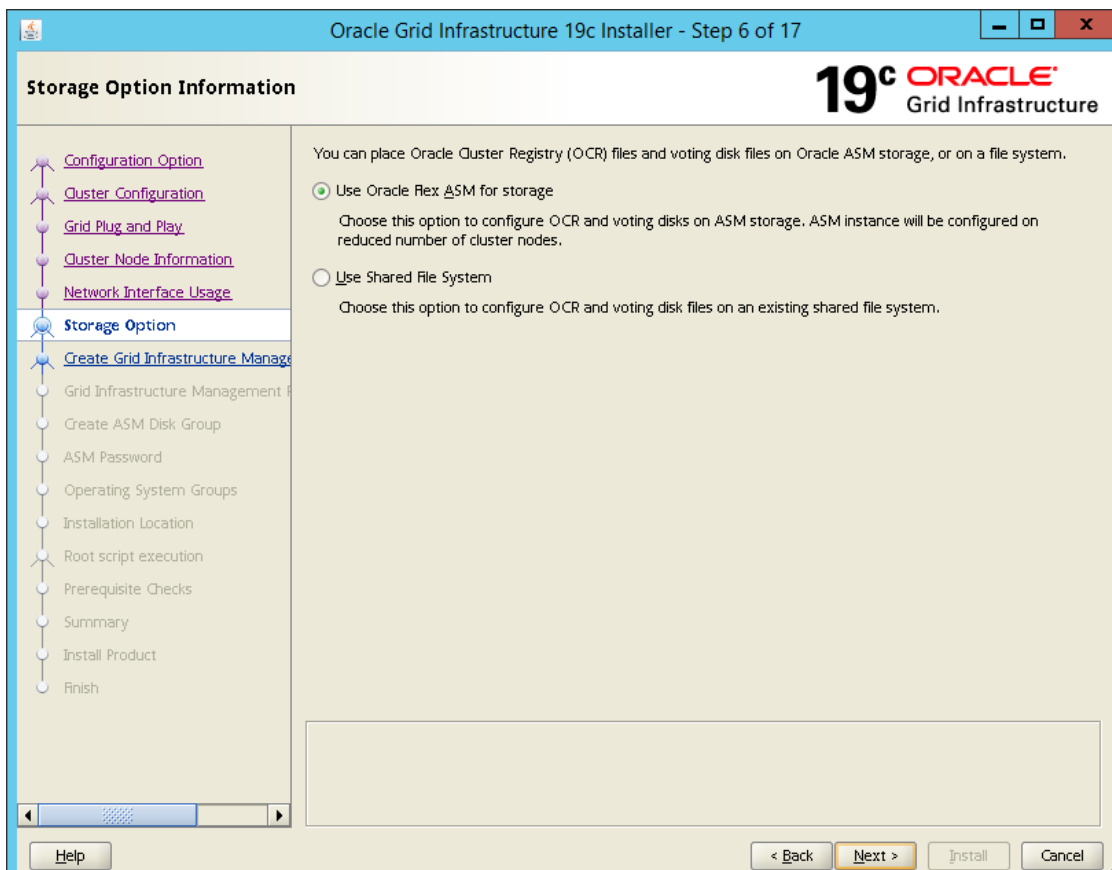
Com os 3 nodes aparecendo na configuração, clique em "SSH connectivity..."



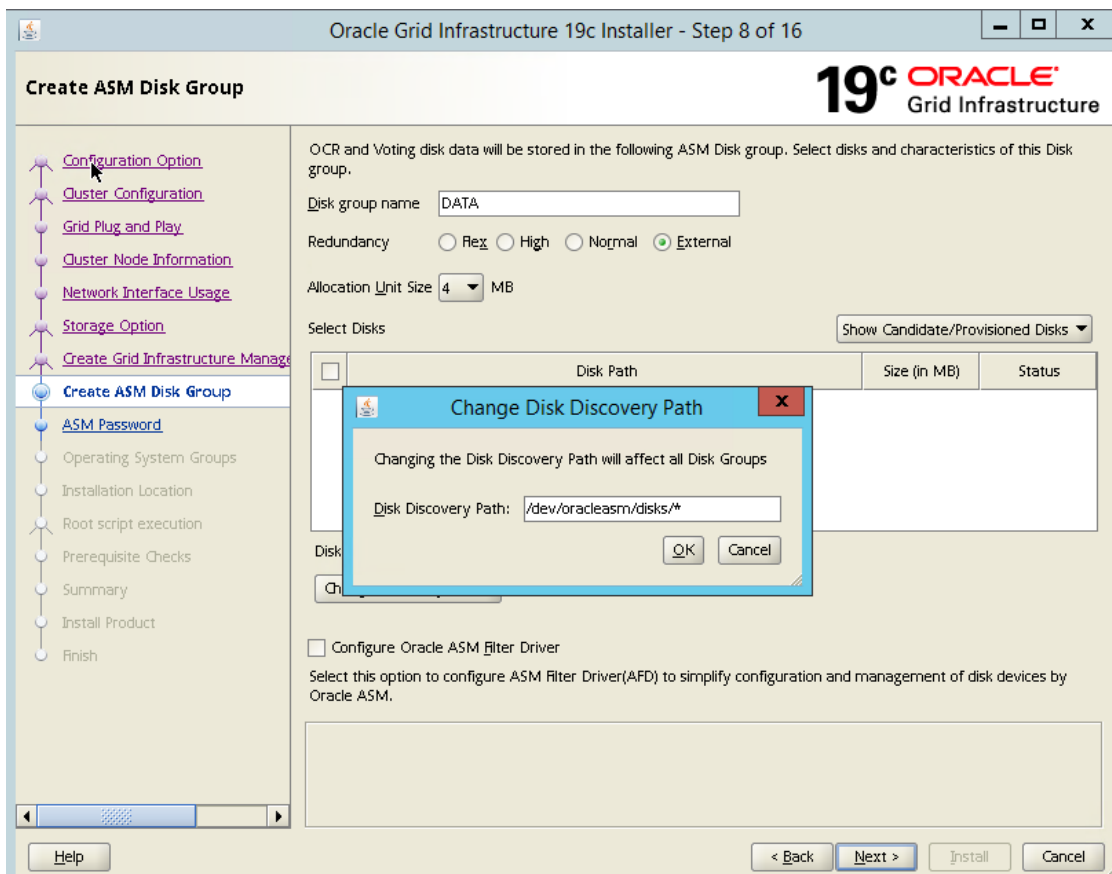
Quanto as interfaces de rede:

- ens161 – rede de backup, não devemos utiliza-la;
- ens192 – rede pública, utilizamos como publica;
- ens193 – aquela que criamos o arquivo de configuração, do interconnect, utilizada com "ASM & Private";
- ens224 – rede branca, não devemos utiliza-la.

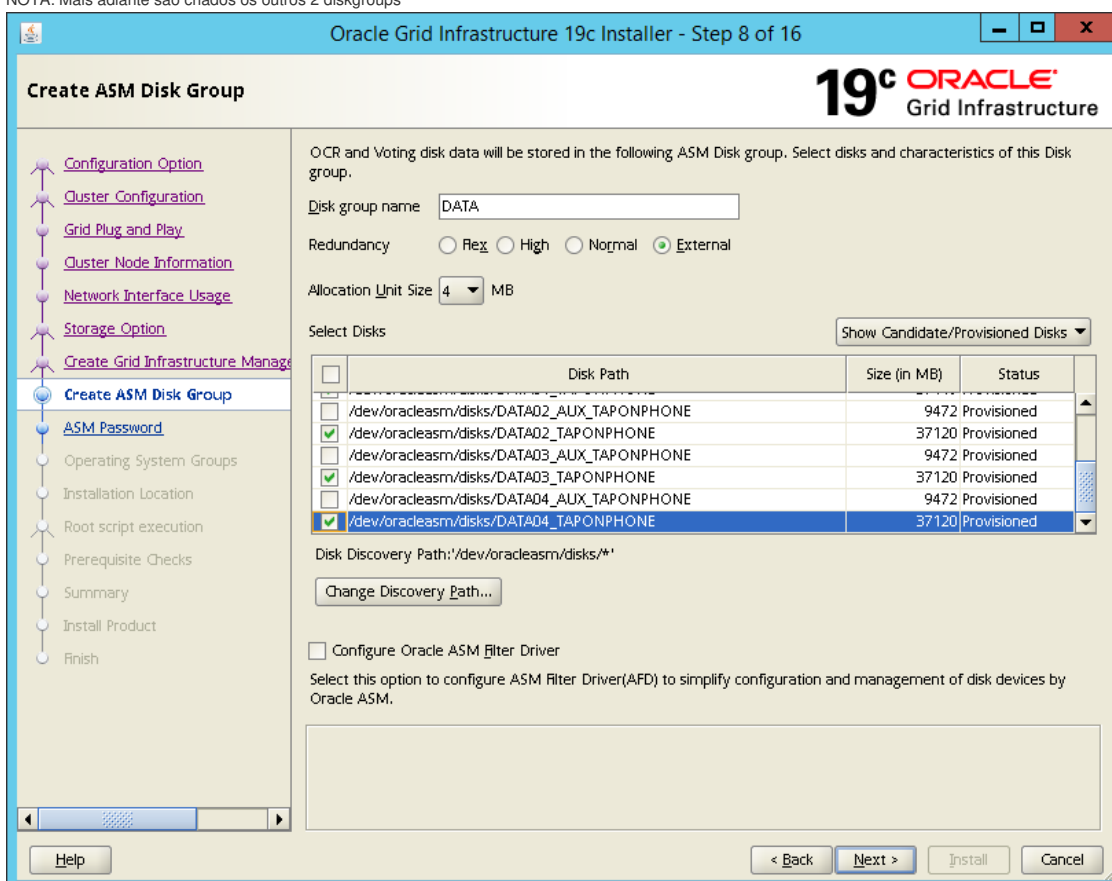




Quanto aos Diskgroups, necessário marcar a opção "external" e alterar o discovery path: /dev/oracleasm/disks/*



Note que todos os discos que criamos logo no começo do procedimento, agora aparecem. Para seguir com o install, vou utilizar só os discos que farão parte do +DATA (DATA_TAPONPHONE)
 NOTA: Mais adiante são criados os outros 2 diskgroups



Utilizamos uma das senhas padrão:

Oracle Grid Infrastructure 19c Installer - Step 11 of 18

19^c ORACLE[®]

Grid Infrastructure

Specify Management Options

Configuration Option

Cluster Configuration

Grid Plug and Play

Cluster Node Information

Network Interface Usage

Storage Option

Create Grid Infrastructure Manag

Create ASM Disk Group

ASM Password

Failure Isolation

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

You can configure to have this instance of Oracle Grid Infrastructure and Oracle Automatic Storage Management to be managed by Enterprise Manager Cloud Control. Specify the details of the Cloud Control configuration to perform the registration.

☐ Register with Enterprise Manager (EM) Cloud Control

O_{MS} host:

O_{MS} port:

EM Admin User Name:

EM Admin Password:

Help

< Back

Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 12 of 18

19^c ORACLE[®]

Grid Infrastructure

Privileged Operating System Groups

Configuration Option

Cluster Configuration

Grid Plug and Play

Cluster Node Information

Network Interface Usage

Storage Option

Create Grid Infrastructure Manag

Create ASM Disk Group

ASM Password

Failure Isolation

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Select the name of the operating system group, that you want to use for operating system authentication to Oracle Automatic Storage Management.

Oracle ASM Administrator (OSASM) Group

Oracle ASM DBA (OSDBA for ASM) Group

Oracle ASM Operator (OSOPER for ASM) Group (Optional)

Help

< Back

Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 13 of 18

19c ORACLE

Grid Infrastructure

Specify Installation Location

Configuration Option

Cluster Configuration

Grid Plug and Play

Cluster Node Information

Network Interface Usage

Storage Option

Create Grid Infrastructure Manage

Create ASM Disk Group

ASM Password

Failure Isolation

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Specify the Oracle base. The Oracle base directory for the Oracle Grid Infrastructure installation is the location where diagnostic and administrative logs, and other logs associated with Oracle ASM and Oracle Clusterware are stored. This location would also contain files pertaining to the configuration of Oracle Clusterware.

Oracle base:

This software directory is the Oracle Grid Infrastructure home directory.

Software location: /u01/app/19.21.0/grid

Help

< Back

Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 14 of 19

19c ORACLE

Grid Infrastructure

Create Inventory

Configuration Option

Cluster Configuration

Grid Plug and Play

Cluster Node Information

Network Interface Usage

Storage Option

Create Grid Infrastructure Manage

Create ASM Disk Group

ASM Password

Failure Isolation

Management Options

Operating System Groups

Installation Location

Create Inventory

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

You are starting your first installation on this host. Specify a directory for installation metadata files (for example, install log files). This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.

Inventory Directory:

Members of the following operating system group (the primary group) will have write permission to the inventory directory (oraInventory).

oraInventory Group Name: oinstall

Help

< Back

Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 15 of 19

Root script execution configuration

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password :

☐ Use sudo

Program path :

User name :

Password :

Ajustar todos os parâmetros errados:

Oracle Grid Infrastructure 19c Installer - Step 16 of 19

Perform Prerequisite Checks

Verification Result

Some of the minimum requirements for installation are not completed. Review and fix the issues listed in the following table, and recheck the system.

☐ Ignore All

Checks	Status	Fixable
Domain Sockets	Warning	No
RPM Package Manager database	Info	No
zeroconf check	Warning	Yes
Kernel retpoline support	Warning	No
(DNS/NIS name service)		
DNS/NIS name service	Failed	No

This task verifies Clusterware domain sockets ([more details](#))

Check Failed on Nodes: [gncashtl08922, gncashtl08921, gncashtl08920]

NOTA: Esqueci de capturar a última tela, porém é de confirmação.

Execução dos scripts como root:

```
[root@GNCASHTL08920 .oracle]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
```

Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.

```
[root@GNCASHTL08921 .oracle]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
```

Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.

```
[root@GNCASHTL08922 .oracle]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
```

Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.

```
[root@GNCASHTL08920 .oracle]# /u01/app/19.21.0/grid/root.sh
Performing root user operation.
```

The following environment variables are set as:
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/19.21.0/grid

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.21.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid/crsdata/gncashtl08920/crsconfig/rootcrs_gncashtl08920_2023-12-05_06-16-45PM.log
2023/12/05 18:16:57 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/12/05 18:16:57 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/12/05 18:16:57 CLSRSC-363: User ignored prerequisites during installation
2023/12/05 18:16:57 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/12/05 18:16:58 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/12/05 18:16:59 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/12/05 18:16:59 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/12/05 18:16:59 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/12/05 18:17:11 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/12/05 18:17:16 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/12/05 18:17:22 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/12/05 18:17:29 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/12/05 18:17:29 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/12/05 18:17:33 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/12/05 18:17:33 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2023/12/05 18:17:55 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/12/05 18:18:00 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/12/05 18:18:05 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/12/05 18:18:09 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
Redirecting to /bin/systemctl restart rsyslog.service

ASM has been created and started successfully.

[DBT-30001] Disk groups created successfully. Check /u01/app/grid/cfgtoollogs/asmca/asmca-231205PM071839.log for details.

2023/12/05 18:19:49 CLSRSC-482: Running command: '/u01/app/19.21.0/grid/bin/ocrconfig -upgrade grid oinstall'
CRS-4256: Updating the profile

```

Successful addition of voting disk fa03f93d05764f8cbfe5c86140b10559.
Successfully replaced voting disk group with +DATA_TAPONPHONE.
CRS-4256: Updating the profile
CRS-4266: Voting file(s) successfully replaced
## STATE   File Universal Id               File Name Disk group
--  -----
1. ONLINE  fa03f93d05764f8cbfe5c86140b10559 (/dev/oracleasm/disks/DATA01_TAPONPHONE) [DATA_TAPONPHONE]
Located 1 voting disk(s).
2023/12/05 18:21:09 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/12/05 18:22:14 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/12/05 18:22:14 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/12/05 18:23:32 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/12/05 18:23:55 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
[root@GNCASHTL08920 .oracle]#

[root@GNCASHTL08921 .oracle]# /u01/app/19.21.0/grid/root.sh
Performing root user operation.

The following environment variables are set as:
  ORACLE_OWNER= grid
  ORACLE_HOME=  /u01/app/19.21.0/grid

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.


Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.21.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
  /u01/app/grid/crsdata/gncashtl08921/crsconfig/rootcrs_gncashtl08921_2023-12-05_06-24-47PM.log
2023/12/05 18:24:53 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/12/05 18:24:54 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/12/05 18:24:54 CLSRSC-363: User ignored prerequisites during installation
2023/12/05 18:24:54 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/12/05 18:24:55 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/12/05 18:24:55 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/12/05 18:24:55 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/12/05 18:24:55 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/12/05 18:24:56 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/12/05 18:24:56 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/12/05 18:25:04 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/12/05 18:25:04 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/12/05 18:25:05 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/12/05 18:25:05 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2023/12/05 18:25:20 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/12/05 18:25:23 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/12/05 18:25:25 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/12/05 18:25:26 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/12/05 18:25:27 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
Redirecting to /bin/systemctl restart rsyslog.service
2023/12/05 18:25:35 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/12/05 18:26:20 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/12/05 18:26:20 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/12/05 18:26:30 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/12/05 18:26:35 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded

[root@GNCASHTL08922 .oracle]# /u01/app/19.21.0/grid/root.sh
Performing root user operation.

The following environment variables are set as:
  ORACLE_OWNER= grid
  ORACLE_HOME=  /u01/app/19.21.0/grid

```

```

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19.21.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
/u01/app/grid/crsdata/gncashtl08922/crsconfig/rootcrs_gncashtl08922_2023-12-05_06-27-06PM.log
2023/12/05 18:27:16 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2023/12/05 18:27:16 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2023/12/05 18:27:16 CLSRSC-363: User ignored prerequisites during installation
2023/12/05 18:27:16 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2023/12/05 18:27:17 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2023/12/05 18:27:17 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2023/12/05 18:27:17 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2023/12/05 18:27:18 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2023/12/05 18:27:19 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2023/12/05 18:27:19 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2023/12/05 18:27:27 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2023/12/05 18:27:27 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2023/12/05 18:27:28 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2023/12/05 18:27:28 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2023/12/05 18:27:43 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2023/12/05 18:27:46 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2023/12/05 18:27:47 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2023/12/05 18:27:48 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2023/12/05 18:27:49 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.
Redirecting to /bin/systemctl restart rsyslog.service
2023/12/05 18:27:57 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2023/12/05 18:28:42 CLSRSC-343: Successfully started Oracle Clusterware stack
2023/12/05 18:28:42 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2023/12/05 18:28:52 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2023/12/05 18:28:57 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded

```

Install Oracle Database

Aplicando o patch nos binários:

```

[oracle@GNCASHTL08920 ~]$ . db_profile
[oracle@GNCASHTL08920 ~]$ xauth add GNCASHTL08920/unix:11 MIT-MAGIC-COOKIE-1 4dd88499d951042189dc05e61292ca1c
xauth: file /home/oracle/.Xauthority does not exist
[oracle@GNCASHTL08920 ~]$ xauth add GNCASHTL08920/unix:11 MIT-MAGIC-COOKIE-1 4dd88499d951042189dc05e61292ca1c
[oracle@GNCASHTL08920 ~]$ export CV_ASSUME_DISTID=OEL8.1
[oracle@GNCASHTL08920 ~]$ cd $ORACLE_HOME
[oracle@GNCASHTL08920 dbhome_1]$ pwd
/u01/app/oracle/product/19.21.0/dbhome_1
[oracle@GNCASHTL08920 dbhome_1]$ ./runInstaller -silent -applyRUR /u02/oracle/patches/30159782/
Preparing the home to patch...
Applying the patch /u02/oracle/patches/30159782/...
Successfully applied the patch.
The log can be found at: /u01/app/orainventory/logs/InstallActions2023-12-06_09-38-17AM/installerPatchActions_2023-12-06_09-38-17AM.log
Launching Oracle Database Setup Wizard...

[FATAL] [INS-35071] Global database name cannot be left blank.
CAUSE: The Global database name was left blank.
ACTION: Specify a value for the Global database name.

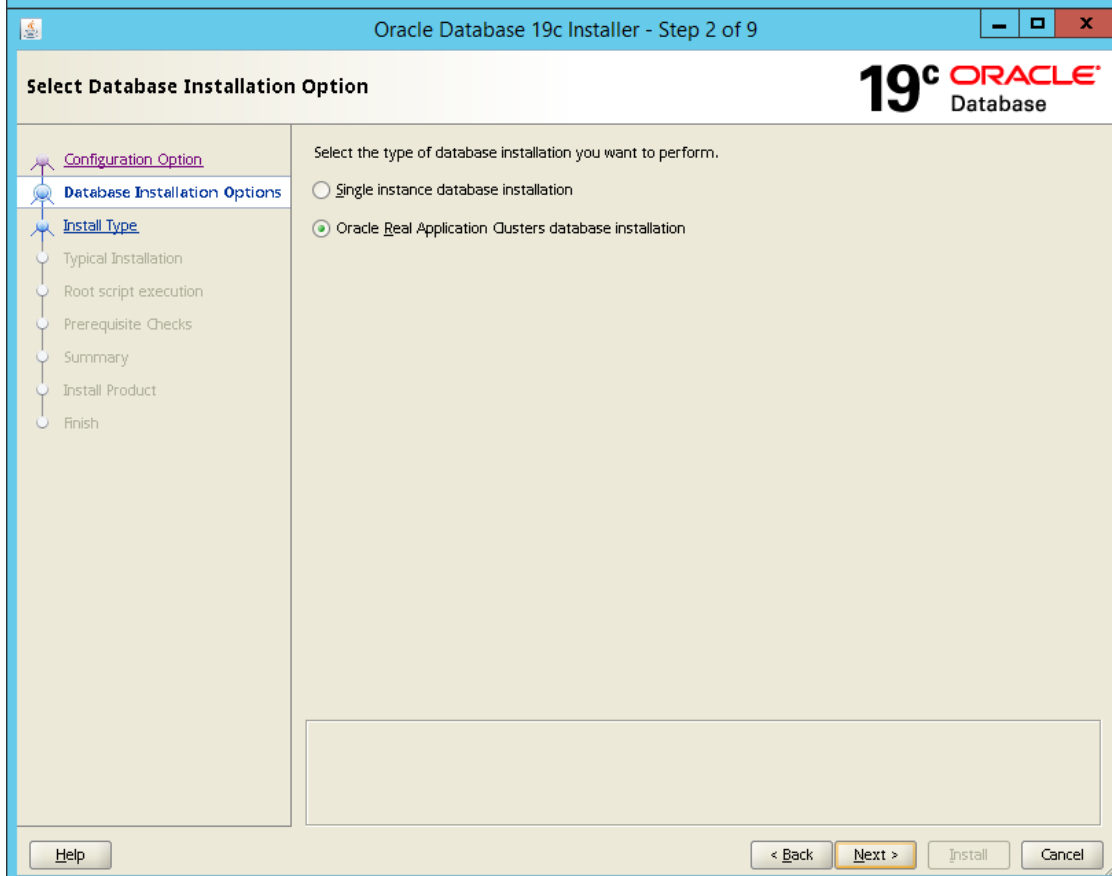
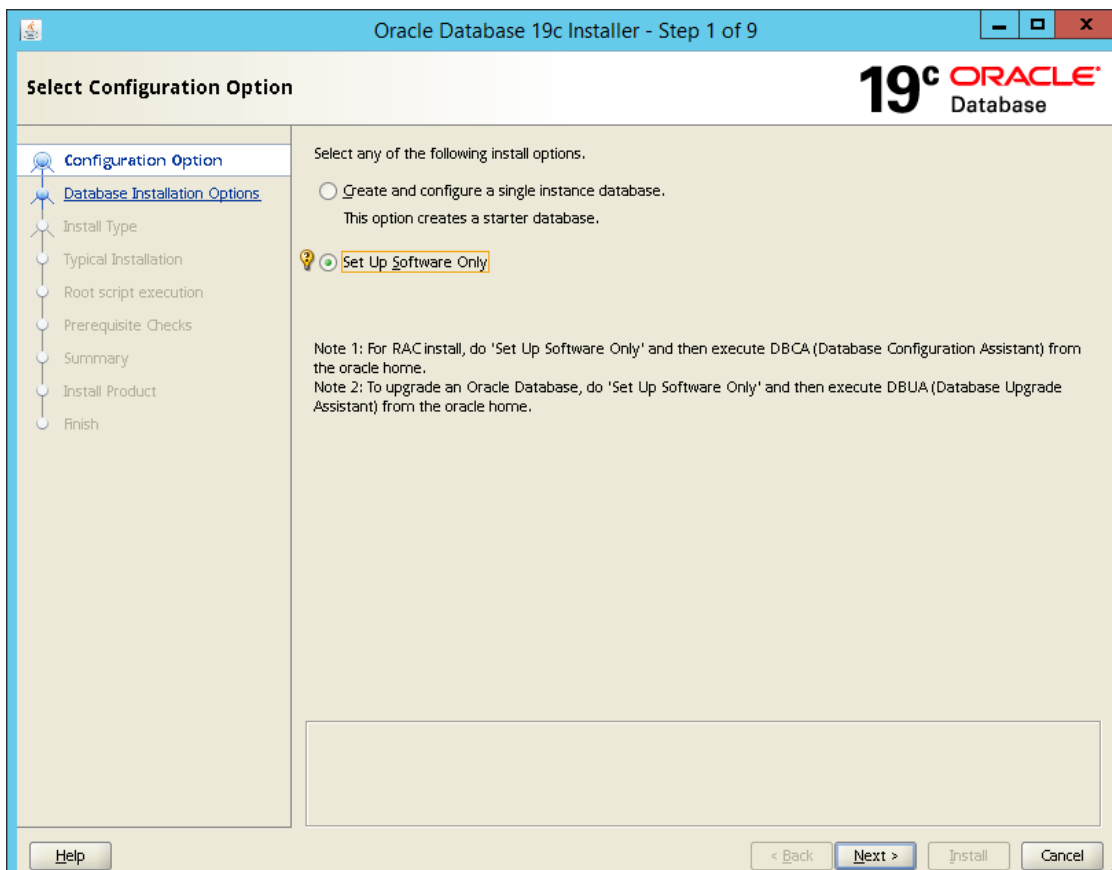
```

Install efetiv:

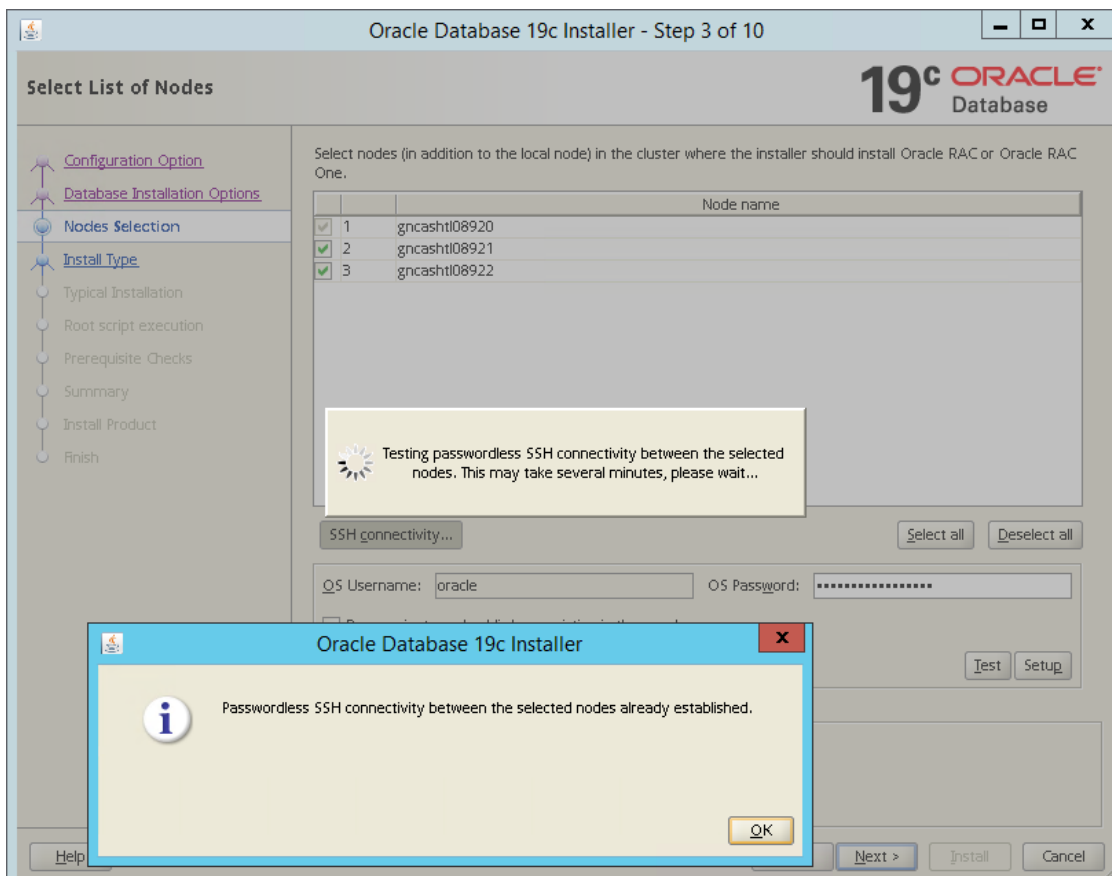
```

[oracle@GNCASHTL08920 dbhome_1]$ export DISPLAY=localhost:11.0
[oracle@GNCASHTL08920 dbhome_1]$ ./runInstaller

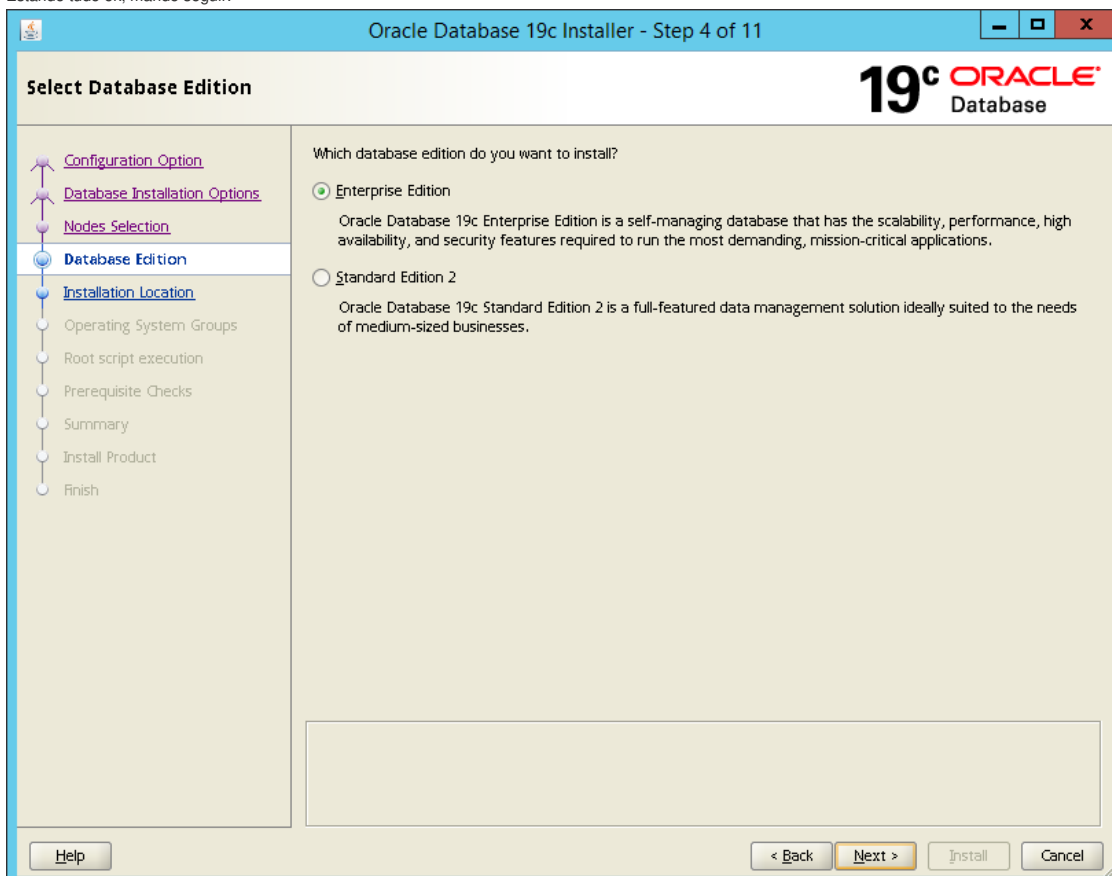
```

Diferente do Grid, no oracle, os 3 nós já aparecem, clique em "SSH Connectivity" coloque a senha do oracle e clique em "Test":



Estando tudo ok, mande seguir.



Oracle Database 19c Installer - Step 5 of 11

19^c ORACLE[®]
Database

Specify Installation Location

[Configuration Option](#)
[Database Installation Options](#)
[Nodes Selection](#)
[Database Edition](#)
[Installation Location](#)
[Operating System Groups](#)
[Root script execution](#)
[Prerequisite Checks](#)
[Summary](#)
[Install Product](#)
[Finish](#)

Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.

Oracle base:

This software directory is the Oracle Database home directory.

Software location: /u01/app/oracle/product/19.21.0/dbhome_1

Oracle Database 19c Installer - Step 6 of 11

19^c ORACLE[®]
Database

Privileged Operating System groups

[Configuration Option](#)
[Database Installation Options](#)
[Nodes Selection](#)
[Database Edition](#)
[Installation Location](#)
[Operating System Groups](#)
[Root script execution](#)
[Prerequisite Checks](#)
[Summary](#)
[Install Product](#)
[Finish](#)

SYS privileges are required to create a database using operating system (OS) authentication. Membership in OS Groups grants the corresponding SYS privilege, eg. membership in OSDBA grants the SYSDBA privilege.

Database Administrator (OSDBA) group:

Database Operator (OSOPER) group (Optional):

Database Backup and Recovery (OSBACKUPDBA) group:

Data Guard administrative (OSDGDBA) group:

Encryption Key Management administrative (OSKMDBA) group:

Real Application Cluster administrative (OSRACDBA) group:

Oracle Database 19c Installer - Step 7 of 11

19c

ORACLE

Database

Root script execution configuration

Configuration Option

Database Installation Options

Nodes Selection

Database Edition

Installation Location

Operating System Groups

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

Use "root" user credential

Password :

Use sudo

Program path : /usr/bin/sudo

Browse...

User name : oracle

Password :

Help

< Back

Next >

Install

Cancel

Conforme informado anteriormente, o RedHat 8 não usa NTP:

Oracle Database 19c Installer - Step 8 of 11

19c

ORACLE

Database

Perform Prerequisite Checks

Configuration Option

Database Installation Options

Nodes Selection

Database Edition

Installation Location

Operating System Groups

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Verification Result

Some of the minimum requirements for installation are not completed. Review and fix the issues listed in the following table, and recheck the system.

Check Again

Fix & Check Again

Show Failed

All Nodes

Ignore All

Checks	Status	Fixable
Checks		
[Clock Synchronization]		
Clock Synchronization	Ignored	No
[Network Time Protocol (NTP)]		
Network Time Protocol (NTP)	Ignored	No
/etc/ntp.conf	Ignored	No
/var/run/ntpd.pid	Ignored	No
Daemon 'ntpd'	Ignored	No
[Single Client Access Name (SCAN)]		
Single Client Access Name (SCAN)	Ignored	No
[DNS/NIS name service 'vip-tapophone-db-oracle']		
DNS/NIS name service 'vip-tapophone-db-oracle'	Ignored	No

This test checks the Oracle Cluster Time Synchronization Services across the cluster nodes. [\(more details\)](#)

Check Failed on Nodes: [gncashtl08922, gncashtl08921, gncashtl08920]

Help

< Back

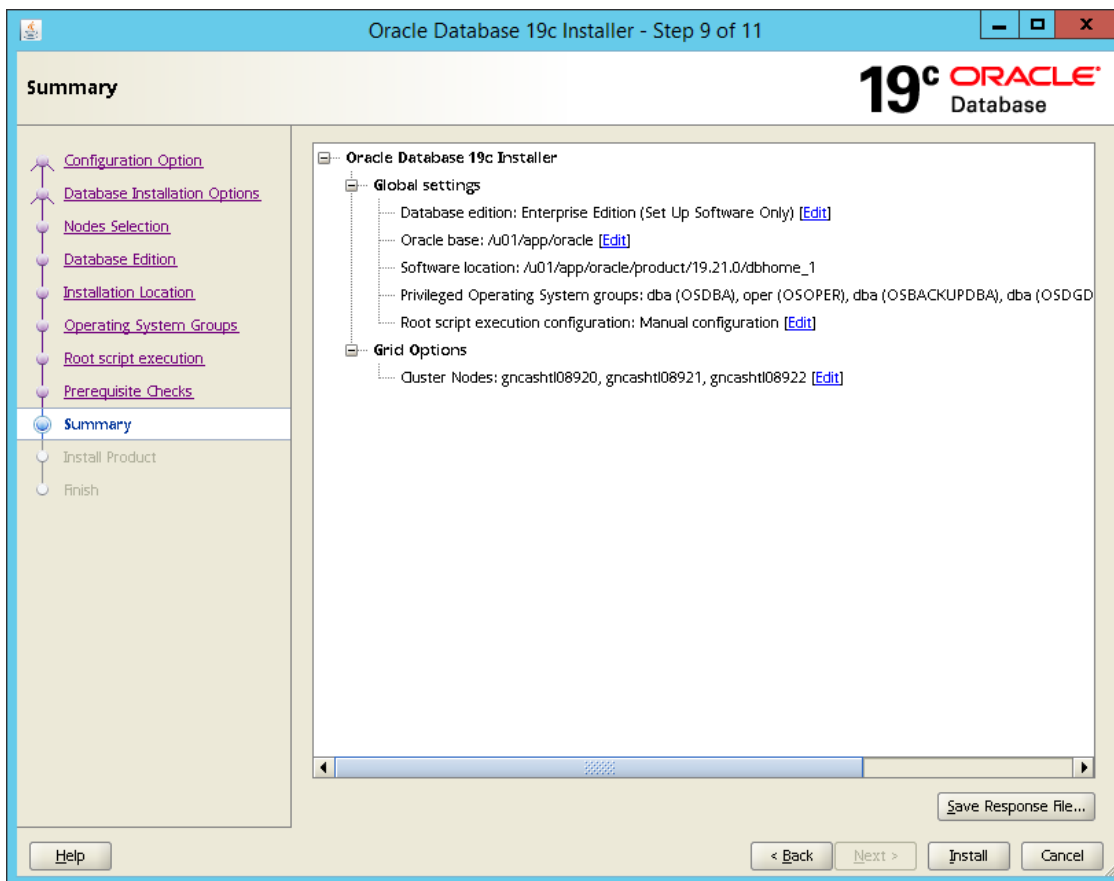
Next >

Install

Cancel

Core Assets/Documentation/GETNET -
DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

44



Script com root:

```
[root@GNCASHTL08920 .oracle]# /u01/app/oracle/product/19.21.0/dbhome_1/root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.21.0/dbhome_1
```

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.

```
[root@GNCASHTL08921 .oracle]# /u01/app/oracle/product/19.21.0/dbhome_1/root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.21.0/dbhome_1
```

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.

```
[root@GNCASHTL08922 .oracle]# /u01/app/oracle/product/19.21.0/dbhome_1/root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.21.0/dbhome_1
```

Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

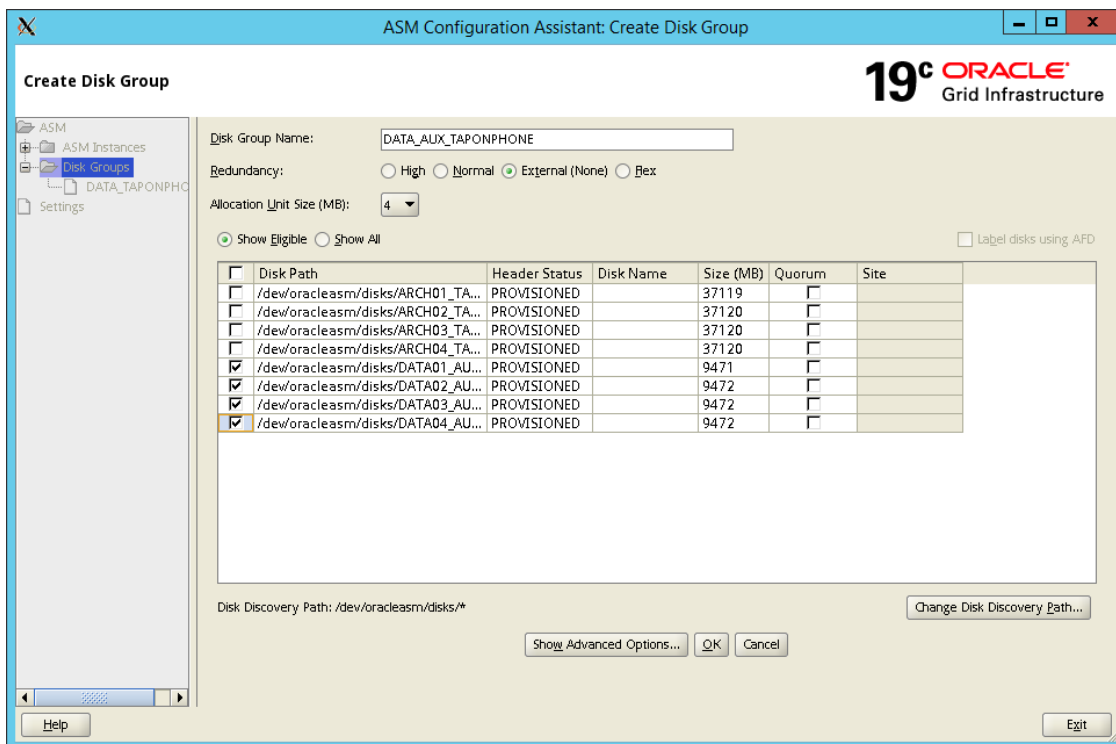
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.

asmca

Criação dos Diskgroups faltantes com user Grid.

```
[grid@GNCASHTL08920 ~]$ . grid_profile
[grid@GNCASHTL08920 ~]$ xauth add GNCASHTL08920/unix:11 MIT-MAGIC-COOKIE-1 4dd88499d951042189dc05e61292ca1c
[grid@GNCASHTL08920 ~]$ export CV_ASSUME_DISTID=OEL8.1
[grid@GNCASHTL08920 ~]$ export DISPLAY=localhost:11.0
[grid@GNCASHTL08920 ~]$ asmca
```

Expanda Disk Groups, clique em "Create...", crie o diskgroup AUX no padrão (DATA_AUX_TAPONPHONE) e escolha os 4 discos referentes a este Disk Group. Lembre-se de marcar o "External".



Repita o processo para o ARCH (ARCH_TAPONPHONE).

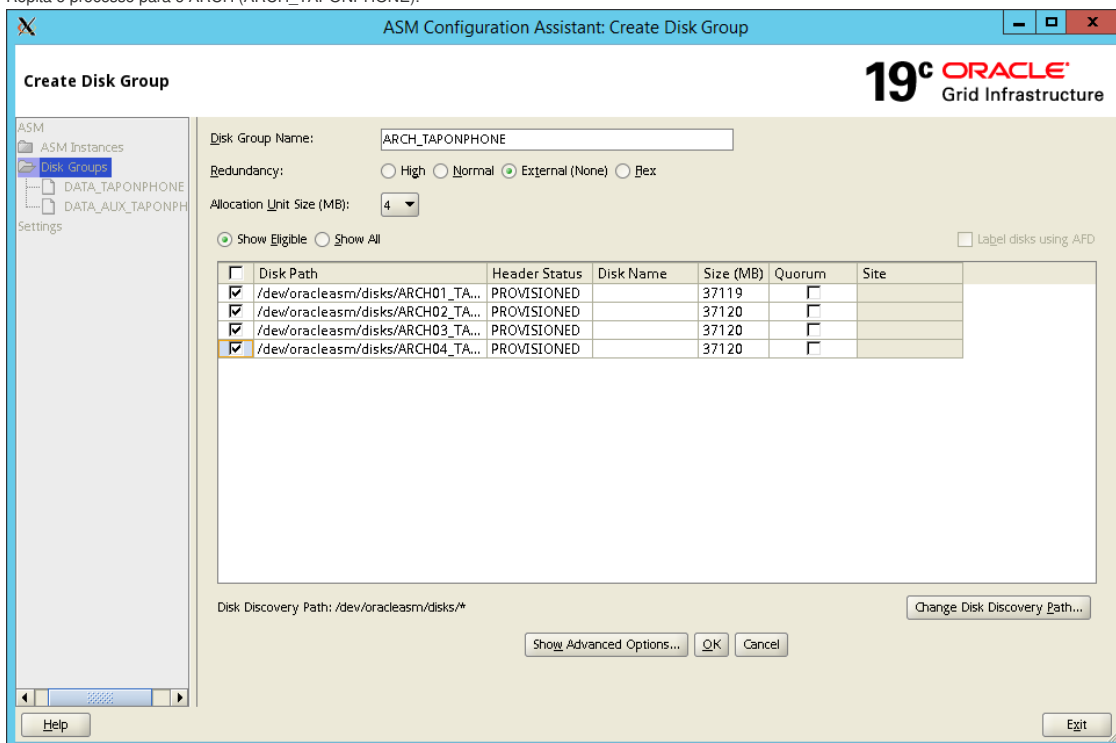
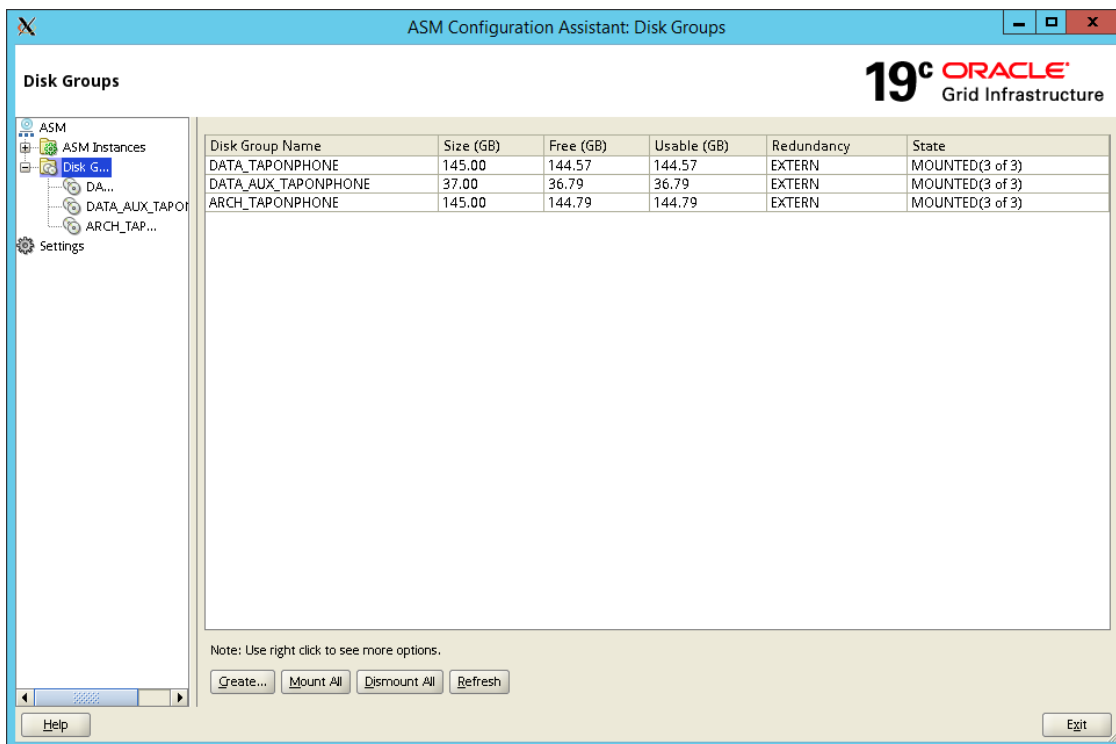


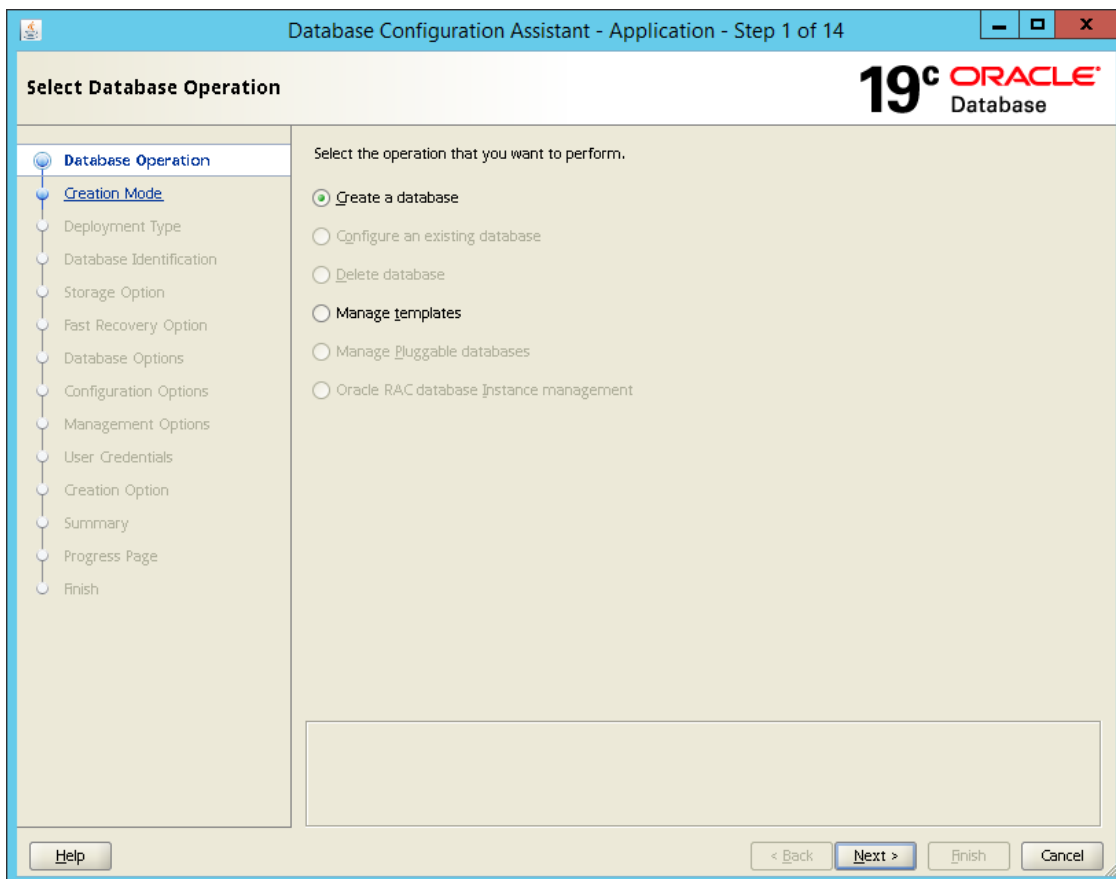
Imagem cm todos os diskgroups criados:



DBCA

Criação do banco de dados com user oracle:

```
[root@GNCASHTL08920 ~]# su - oracle
Last login: Wed Dec 6 09:58:33 -03 2023
[oracle@GNCASHTL08920 ~]$ . db_profile
[oracle@GNCASHTL08920 ~]$ xauth add GNCASHTL08920/unix:11 MIT-MAGIC-COOKIE-1 4dd88499d951042189dc05e61292ca1c
[oracle@GNCASHTL08920 ~]$ export CV_ASSUME_DISTID=OEL8.1
[oracle@GNCASHTL08920 ~]$ export DISPLAY=localhost:11.0
[oracle@GNCASHTL08920 ~]$ dbca
```



Database Configuration Assistant - Create a database - Step 2 of 14

19c ORACLE Database

Select Database Creation Mode

Database Operation
Creation Mode
Deployment Type
Database Identification
Storage Option
Fast Recovery Option
Database Options
Configuration Options
Management Options
User Credentials
Creation Option
Summary
Progress Page
Finish

☐ Typical configuration

Global database name:

Storage type:

Database files location:

Fast Recovery Area (FRA):

Database character set:

Administrative password:

Confirm password:

☒ Create as Container database

Pluggable database name:

☒ Advanced configuration

Database Configuration Assistant - Create a database - Step 3 of 14

19c ORACLE Database

Select Database Deployment Type

Database Operation
Creation Mode
Deployment Type
Database Identification
Storage Option
Fast Recovery Option
Database Options
Configuration Options
Management Options
User Credentials
Creation Option
Summary
Progress Page
Finish

Select the type of database you want to create.

Database type:

Configuration type:

Select a template for your database.

Templates that include datafiles contain pre-created databases. They allow you to create a new database quickly. Use templates without datafiles only when necessary, such as when you need to change attributes like block size that cannot be altered after database creation.

Template name	Include datafiles	Details
☐ Data Warehouse	Yes	View details
☐ Custom Database	No	View details
☒ General Purpose or Transaction Processing	Yes	View details

Template location:

Database Configuration Assistant - Create a database - Step 4 of 16

19^c ORACLE[®]

Database

Select List of Nodes

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Select the nodes on which you want to create the cluster database. The local node "gncashtl08920" should always be selected.

	Node name
☒	1 gncashtl08920
☒	2 gncashtl08921
☒	3 gncashtl08922

Select all

Deselect all

Help

< Back

Next >

Finish

Cancel

NOTA: Veja que o SID não ficará o nome completo, depois é necessário corrigir nos db_profiles

Database Configuration Assistant - Create a database - Step 5 of 16

19^c ORACLE[®]

Database

Specify Database Identification Details

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Provide a unique database identifier information. An Oracle database is uniquely identified by a Global database name, typically of the form "name.domain".

Global database name:

TAPONPHONE

SID Prefix:

TAPONPHO

Service name:

☐ Create as Container database

A Container database can be used for consolidating multiple databases into a single database, and it enables database virtualization. A Container database (CDB) can have zero or more pluggable databases (PDB).

☒ Use local Undo tablespace for PDBs

☐ Create an empty Container database

☒ Create a Container database with one or more PDBs

Number of PDBs:

1

PDB name:

pdb

Help

< Back

Next >

Finish

Cancel

Core Assets/Documentation/GETNET -
DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

50

Database Configuration Assistant - Create 'TAPONPHO' database - Step 6 of 16

19c ORACLE Database

Select Database Storage Option

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Use template file for database storage attributes

Storage type and location for database files will be picked up from the specified template (General Purpose or Transaction Processing).

Use following for the database storage attributes

All the database files will be put at the specified location below. You can customize the name and location of each datafile in the subsequent screen.

Database files storage type: Automatic Storage Management (ASM)

Database files location: +DATA_TAPONPHONE/{DB_UNIQUE_NAME} Browse...

Oracle Managed files option will enable Oracle to automatically generate the names of the datafiles for simplified database management.

☒ Use Oracle-Managed Files (OMF)

Multiplex redo logs and control files...

File location variables...

Help

< Back

Next >

Finish

Cancel

Configuração de Archives será feita posteriormente.

Database Configuration Assistant - Create 'TAPONPHO' database - Step 7 of 16

19c ORACLE Database

Select Fast Recovery Option

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Choose the recovery options for the database.

Specify Fast Recovery Area

Recovery files storage type: Automatic Storage Management (ASM)

Fast Recovery Area: +DATA_TAPONPHONE Browse...

Fast Recovery Area size: 8256 MB

☐ Enable archiving

Edit archive mode parameters...

Help

< Back

Next >

Finish

Cancel

Core Assets/Documentation/GETNET - DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

51

Database Configuration Assistant - Create 'TAPONPHO' database - Step 8 of 16

19c ORACLE Database

Select Oracle Data Vault Config Option

- Database Operation
- Creation Mode
- Deployment Type
- Nodes Selection
- Database Identification
- Storage Option
- Fast Recovery Option
- Data Vault Option**
- Configuration Options
- Management Options
- User Credentials
- Creation Option
- Prerequisite Checks
- Summary
- Progress Page
- Finish

☐ Configure Oracle Database Vault
Database Vault owner:
Password: Confirm password:
☐ Create a separate account manager
Account manager:
Password: Confirm password:
☐ Configure Oracle Label Security
☐ Configure Oracle Label Security with OJD

NOTA: Na tela seguinte vamos navegar por todas as abas:

Database Configuration Assistant - Create 'TAPONPHO' database - Step 9 of 16

19c ORACLE Database

Specify Configuration Options

- Database Operation
- Creation Mode
- Deployment Type
- Nodes Selection
- Database Identification
- Storage Option
- Fast Recovery Option
- Data Vault Option
- Configuration Options**
- Management Options
- User Credentials
- Creation Option
- Prerequisite Checks
- Summary
- Progress Page
- Finish

Memory
Sizing
Character sets
Connection mode
Sample schemas

☒ Use Automatic Shared Memory Management
SGA size: MB
PGA size: MB
☐ Use Manual Shared Memory Management
Shared pool size: MB
Buffer cache size: MB
Java pool size: MB
Large pool size: MB
PGA size: MB
Total memory for database 0 MB
☐ Use Automatic Memory Management
Memory target: MB 39%

Database Configuration Assistant - Create 'TAPONPHO' database - Step 9 of 16

Specify Configuration Options

19^c
ORACLE[®]
Database

Memory
Sizing
Character sets
Connection mode
Sample schemas

A block is the smallest unit of storage for allocation and for I/O. It cannot be changed once the database is created.

Block size:
8192
BYTES

Specify the maximum number of operating system user processes that can be simultaneously connected to this database. The value of this parameter includes the user processes and the Oracle background processes.

Processes:
2500

Database Operation
Creation Mode
Deployment Type
Nodes Selection
Database Identification
Storage Option
Fast Recovery Option
Data Vault Option
Configuration Options
Management Options
User Credentials
Creation Option
Prerequisite Checks
Summary
Progress Page
Finish

Help

< Back
Next >
Finish
Cancel

NOTA: Conforme indicação do Rodrigo Ignasi, através da evidência abaixo retirada em HTI, devemos usar o charset abaixo:

HTI - VIP - RAC - TAP ON PHONE

Planilha
Query Builder

1
2

SELECT * FROM V\$NLS_PARAMETERS;

Resultado da Consulta

SQL
Todas as Linhas Extraídas: 19 em 0,113 segundos

PARAMETER	VALUE	CON_ID
1 NLS_LANGUAGE	BRAZILIAN PORTUGUESE	1
2 NLS_TERRITORY	BRAZIL	1
3 NLS_CURRENCY	R\$	1
4 NLS_ISO_CURRENCY	BRAZIL	1
5 NLS_NUMERIC_CHARACTERS	,.	1
6 NLS_CALENDAR	GREGORIAN	1
7 NLS_DATE_FORMAT	DD/MM/RR	1
8 NLS_DATE_LANGUAGE	BRAZILIAN PORTUGUESE	1
9 NLS_CHARACTERSET	AL32UTF8	1
10 NLS_SORT	WEST_EUROPEAN	1
11 NLS_TIME_FORMAT	HH24:MI:SSXFF	1
12 NLS_TIMESTAMP_FORMAT	DD/MM/RR HH24:MI:SSXFF	1
13 NLS_TIME_TZ_FORMAT	HH24:MI:SSXFF TZR	1
14 NLS_TIMESTAMP_TZ_FORMAT	DD/MM/RR HH24:MI:SSXFF TZR	1
15 NLS_DUAL_CURRENCY	Cr\$	1
16 NLS_NCHAR_CHARACTERSET	AL16UTF16	1
17 NLS_COMP	BINARY	1
18 NLS_LENGTH_SEMANTICS	BYTE	1
19 NLS_NCHAR_CONV_EXCP	FALSE	1

Core Assets/Documentation/GETNET - DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

53

Database Configuration Assistant - Create 'TAPONPHO' database - Step 9 of 16

Specify Configuration Options

19c ORACLE Database

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Memory

Sizing

Character sets

Connection mode

Sample schemas

The database character set determines how character data is stored in the database.

☒ Use Unicode (AL32UTF8)

Setting character set to Unicode (AL32UTF8) enables you to store multiple language groups.

☐ Use OS character set (WE8MSWIN1252)

Character set is based on the language setting of this operating system.

☐ Choose from the list of character sets

Database character set:

☒ Show recommended character sets only

National character set:

Default language:

Default territory:

Help

< Back

Next >

Finish

Cancel

Database Configuration Assistant - Create 'TAPONPHO' database - Step 10 of 16

Specify Management Options

19c ORACLE Database

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Specify the management options for the database.

☐ Run Cluster Verification Utility (CVU) checks periodically

☐ Configure Enterprise Manager (EM) database express

EM database express port:

☐ Register with Enterprise Manager (EM) cloud control

OMS host:

OMS port:

EM admin username:

EM admin password:

ASMSNMP user password:

Help

< Back

Next >

Finish

Cancel

Mais uma vez, uma das senhas padrões:

Core Assets/Documentation/GETNET -
DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

54

Database Configuration Assistant - Create 'TAPONPHO' database - Step 11 of 16

19c ORACLE Database

Specify Database User Credentials

- Database Operation
- Creation Mode
- Deployment Type
- Nodes Selection
- Database Identification
- Storage Option
- Fast Recovery Option
- Data Vault Option
- Configuration Options
- Management Options
- User Credentials**
- Creation Option
- Prerequisite Checks
- Summary
- Progress Page
- Finish

You must specify passwords for the following user accounts in the new database for security reasons.

☐ Use different administrative passwords

	Password	Confirm password
SYS		
SYSTEM		

☒ Use the same administrative password for all accounts

Password:
Confirm password:

Messages:

Warning: [DBT-06208] The 'ADMIN' password entered does not conform to the Oracle recommended standards.

Help
< Back
Next >
Finish
Cancel

Database Configuration Assistant - Create 'TAPONPHO' database - Step 12 of 16

19c ORACLE Database

Select Database Creation Option

- Database Operation
- Creation Mode
- Deployment Type
- Nodes Selection
- Database Identification
- Storage Option
- Fast Recovery Option
- Data Vault Option
- Configuration Options
- Management Options
- User Credentials
- Creation Option**
- Prerequisite Checks
- Summary
- Progress Page
- Finish

Select the database creation options.

☒ Create database

Specify the SQL scripts you want to run after the database is created. The scripts are run in the order listed below.

Post DB creation scripts:

☐ Save as a database template

Template name:

Template location:

Description:

☐ Generate database creation scripts

Destination directory:

Following advanced configuration options can be used to configure initialization parameters and customize database storage locations.

Help
< Back
Next >
Finish
Cancel

Database Configuration Assistant - Create 'TAPONPHO' database - Step 13 of 16

Perform Prerequisite Checks

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Verification Result

Some of the minimum requirements for installation are not completed. Review and fix the issues listed in the following table, and recheck the system.

Check Again

Fix & Check Again

Show Failed

All Nodes

Ignore All

Checks	Status	Fixable
Checks		
[Single Client Access Name (SCAN)]		
Single Client Access Name (SCAN)	Ignored	No
[DNS/NIS name service 'vip-taponphone-db-oracle']		
DNS/NIS name service 'vip-taponphone-db-oracle'	Ignored	No

This test verifies the Single Client Access Name configuration. [\(more details\)](#)

Check Failed on Nodes: [gncashtl08922, gncashtl08921, gncashtl08920]

Help

< Back

Next >

Finish

Cancel

Database Configuration Assistant - Create 'TAPONPHO' database - Step 14 of 16

Summary

Database Operation

Creation Mode

Deployment Type

Nodes Selection

Database Identification

Storage Option

Fast Recovery Option

Data Vault Option

Configuration Options

Management Options

User Credentials

Creation Option

Prerequisite Checks

Summary

Progress Page

Finish

Database Configuration Assistant

Global Settings

Global database name: TAPONPHONE

Node List: gncashtl08920,gncashtl08921,gncashtl08922

Configuration type: Oracle Real Application Cluster (RAQ) database - Admin Managed

SID: TAPONPHO

Create as Container database: No

Database Files Storage Type: Automatic Storage Management (ASM)

Memory Configuration Type: Automatic Shared Memory Management

Template name: General Purpose

Initialization Parameters

audit_file_dest: {ORACLE_BASE}/admin/{DB_UNIQUE_NAME}/adump

audit_trail: db

cluster_database: true

compatible: 19.0.0

db_block_size: 8192 BYTES

db_create_file_dest: +DATA_TAPONPHONE/{DB_UNIQUE_NAME}/

db_name: TAPONPHO

db_unique_name: TAPONPHONE

diagnostic_dest: {ORACLE_BASE}

dispatchers: (PROTOCOL=TCP) (SERVICE=TAPONPHOXDB)

family:dw_helper.instance_mode: read-only

instance_number (TAPONPHO1): 1

Save Response File...

Help

< Back

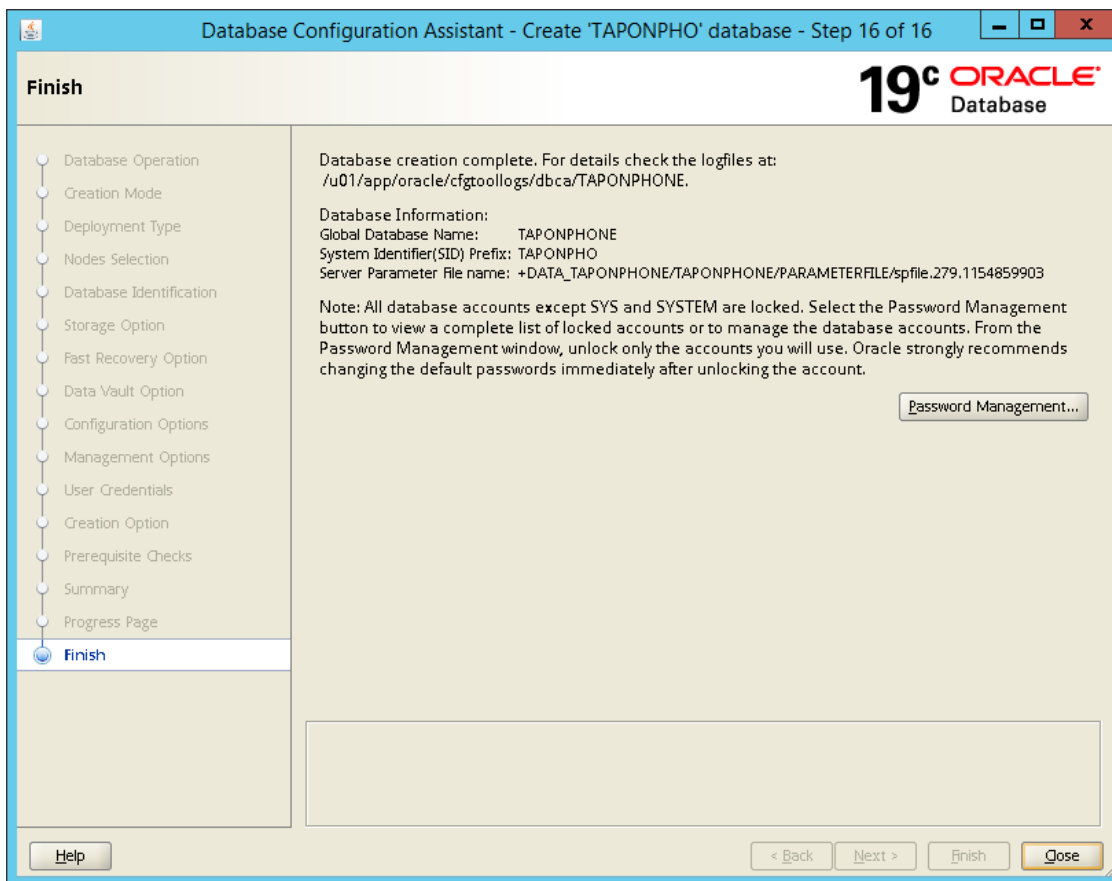
Next >

Finish

Cancel

Core Assets/Documentation/GETNET - DBA
ORAEX Cloud Consulting | GETNET | Jan 30, 2026 - 12:03am

56



PSU 19.21

NOTA: Conforme indiquei o Kernell é Redhat 8.8, sendo assim, apliquei o PSU 19.21.

NOTA2: Foi necessário fazer o mesmo export do GridSetup e runInstaller.

NOTA3: O Opatch já foi tratado no início.

Necessário fazer o unzip dos 3 arquivos de PSU, OJVM e JDK:

```
[grid@GNCASHTL08920 patches]$ ls -la
total 9865892
drwxrwxr-x 6 grid oinstall 4096 Dec 6 10:32 .
drwxrwxr-x 4 grid oinstall 35 Dec 4 12:19 ..
drwxr-x--- 4 grid oinstall 95 Dec 1 2019 30159782
drwxr-xr-x 4 grid oinstall 67 Sep 25 11:41 35638318
drwxr-x--- 8 grid oinstall 159 Oct 4 11:47 35642822
drwxr-xr-x 4 grid oinstall 67 Aug 25 14:22 35648110
-rw-rw---- 1 grid oinstall 3059705302 Feb 24 2020 LINUX.X64_193000_db_home.zip
-rw-rw---- 1 grid oinstall 2889184573 Feb 24 2020 LINUX.X64_193000_grid_home.zip
-rw-r--r-- 1 grid oinstall 692329271 May 29 2023 p30159782_195000OCWRU_Linux-x86-64.zip
-rw-rw---- 1 grid oinstall 121857143 Oct 18 18:31 p35638318_190000_Linux-x86-64.zip
-rw-rw---- 1 grid oinstall 3084439097 Oct 18 18:30 p35642822_190000_Linux-x86-64.zip
-rw-rw---- 1 grid oinstall 127350205 Oct 18 18:31 p35648110_190000_Linux-x86-64.zip
-rw-rw---- 1 grid oinstall 127774864 Oct 18 18:30 p6880880_190000_Linux-x86-64.zip
-rw-rw-r-- 1 grid oinstall 11676 Oct 17 14:17 PatchSearch.xml
```

Aplicando o PSU com user root, costumo sempre fazer no nó 01 antes do nó 02 e 03. Datapatch ao fim do nó 03:

```
[root@GNCASHTL08920 .oracle]# ./home/grid/grid_profile
[root@GNCASHTL08920 .oracle]# cd $ORACLE_HOME/OPatch
[root@GNCASHTL08920 OPatch]# export CV_ASSUME_DISTID=OEL8.1
[root@GNCASHTL08920 OPatch]# ./opatchauto apply /u02/oracle/patches/35642822
```

OPatchauto session is initiated at Wed Dec 6 10:36:01 2023

System initialization log file is /u01/app/19.21.0/grid/cfgtoollogs/opatchautodb/systemconfig2023-12-06_10-36-07AM.log.

Session log file is /u01/app/19.21.0/grid/cfgtoollogs/opatchauto/opatchauto2023-12-06_10-36-28AM.log

The id for this session is NY11

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.21.0/grid

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1

Patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Patch applicability verified successfully on home /u01/app/19.21.0/grid

Executing patch validation checks on home /u01/app/19.21.0/grid

Patch validation checks successfully completed on home /u01/app/19.21.0/grid

Executing patch validation checks on home /u01/app/oracle/product/19.21.0/dbhome_1

Patch validation checks successfully completed on home /u01/app/oracle/product/19.21.0/dbhome_1

Verifying SQL patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1

SQL patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Preparing to bring down database service on home /u01/app/oracle/product/19.21.0/dbhome_1

Successfully prepared home /u01/app/oracle/product/19.21.0/dbhome_1 to bring down database service

Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.21.0/grid

Prepatch operation log file location: /u01/app/grid/crsdata/gncashtl08920/crsconfig/crs_prepatch_apply_inplace_gncashtl08920_2023-12-06_10-38-06AM.log

CRS service brought down successfully on home /u01/app/19.21.0/grid

Performing prepatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1

Prepatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/oracle/product/19.21.0/dbhome_1

Binary patch applied successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Running rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Successfully executed rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Performing postpatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1

Postpatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/19.21.0/grid

Binary patch applied successfully on home /u01/app/19.21.0/grid

Running rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Successfully executed rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.21.0/grid

Postpatch operation log file location: /u01/app/grid/crsdata/gncashtl08920/crsconfig/crs_postpatch_apply_inplace_gncashtl08920_2023-12-06_10-52-36AM.log

CRS service started successfully on home /u01/app/19.21.0/grid

Preparing home /u01/app/oracle/product/19.21.0/dbhome_1 after database service restarted

No step execution required.....

Trying to apply SQL patch on home /u01/app/oracle/product/19.21.0/dbhome_1

No SQL patch operations are required on local node for this home

OPatchAuto successful.

-----Summary-----

Patching is completed successfully. Please find the summary as follows:

Host:gncashtl08920
RAC Home:/u01/app/oracle/product/19.21.0/dbhome_1
Version:19.0.0.0.0
Summary:

==Following patches were SKIPPED:

Patch: /u02/oracle/patches/35642822/35652062
Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/35642822/33575402
Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/35642822/35553096
Reason: This patch is not applicable to this specified target type - "rac_database"

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35642822/35643107
Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_10-40-47AM_1.log

Patch: /u02/oracle/patches/35642822/35655527
Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_10-40-47AM_1.log

Host:gncashtl08920
CRS Home:/u01/app/19.21.0/grid
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35642822/33575402
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_10-46-57AM_1.log

Patch: /u02/oracle/patches/35642822/35553096
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_10-46-57AM_1.log

Patch: /u02/oracle/patches/35642822/35643107
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_10-46-57AM_1.log

Patch: /u02/oracle/patches/35642822/35652062
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_10-46-57AM_1.log

Patch: /u02/oracle/patches/35642822/35655527
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_10-46-57AM_1.log

OPatchauto session completed at Wed Dec 6 10:57:45 2023
Time taken to complete the session 21 minutes, 39 seconds
[root@GNCASHTL08920 OPatch]#

Aplicando PSU no nó 02:

```
[root@GNCASHTL08921 .oracle]# cd $ORACLE_HOME/OPatch
[root@GNCASHTL08921 OPatch]# export CV_ASSUME_DISTID=OEL8.1
[root@GNCASHTL08921 OPatch]# ./opatchauto apply /u02/oracle/patches/35642822
```

OPatchauto session is initiated at Wed Dec 6 10:58:20 2023

System initialization log file is /u01/app/19.21.0/grid/cfgtoollogs/patchautodb/systemconfig2023-12-06_10-58-25AM.log.

Session log file is /u01/app/19.21.0/grid/cfgtoollogs/patchauto/patchauto2023-12-06_10-58-50AM.log
The id for this session is 82UV

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.21.0/grid

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1
Patch applicability verified successfully on home /u01/app/19.21.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Executing patch validation checks on home /u01/app/19.21.0/grid
Patch validation checks successfully completed on home /u01/app/19.21.0/grid

Executing patch validation checks on home /u01/app/oracle/product/19.21.0/dbhome_1
Patch validation checks successfully completed on home /u01/app/oracle/product/19.21.0/dbhome_1

Verifying SQL patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1
SQL patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Preparing to bring down database service on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully prepared home /u01/app/oracle/product/19.21.0/dbhome_1 to bring down database service

Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.21.0/grid
Prepatch operation log file location: /u01/app/grid/crsdata/gncashtl08921/crsconfig/crs_prepatch_apply_inplace_gncashtl08921_2023-12-06_11-00-26AM.log
CRS service brought down successfully on home /u01/app/19.21.0/grid

Performing prepatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Prepatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/oracle/product/19.21.0/dbhome_1
Binary patch applied successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Running rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully executed rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Performing postpatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Postpatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/19.21.0/grid
Binary patch applied successfully on home /u01/app/19.21.0/grid

Running rootadd_rdbms.sh on home /u01/app/19.21.0/grid
Successfully executed rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.21.0/grid
Postpatch operation log file location: /u01/app/grid/crsdata/gncashtl08921/crsconfig/crs_postpatch_apply_inplace_gncashtl08921_2023-12-06_11-14-23AM.log
CRS service started successfully on home /u01/app/19.21.0/grid

Preparing home /u01/app/oracle/product/19.21.0/dbhome_1 after database service restarted
No step execution required.....

Trying to apply SQL patch on home /u01/app/oracle/product/19.21.0/dbhome_1
No SQL patch operations are required on local node for this home

OPatchAuto successful.

-----Summary-----

Patching is completed successfully. Please find the summary as follows:

Host:gncashtl08921
RAC Home:/u01/app/oracle/product/19.21.0/dbhome_1
Version:19.0.0.0.0
Summary:

==Following patches were SKIPPED:

Patch: /u02/oracle/patches/35642822/35652062
Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/35642822/33575402
Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/35642822/35553096
Reason: This patch is not applicable to this specified target type - "rac_database"

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35642822/35643107
Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-02-55AM_1.log

Patch: /u02/oracle/patches/35642822/35655527
Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-02-55AM_1.log

Host:gncashtl08921
CRS Home:/u01/app/19.21.0/grid
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35642822/33575402
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-08-56AM_1.log

Patch: /u02/oracle/patches/35642822/35553096
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-08-56AM_1.log

Patch: /u02/oracle/patches/35642822/35643107
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-08-56AM_1.log

Patch: /u02/oracle/patches/35642822/35652062
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-08-56AM_1.log

Patch: /u02/oracle/patches/35642822/35655527
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-08-56AM_1.log

OPatchauto session completed at Wed Dec 6 11:19:26 2023
Time taken to complete the session 21 minutes, 1 second

Aplicando no nó 03 + Datapatch do PSU

```
[root@GNCASHTL08922 .oracle]# cd $ORACLE_HOME/OPatch
[root@GNCASHTL08922 OPatch]# export CV_ASSUME_DISTID=OEL8.1
[root@GNCASHTL08922 OPatch]# ./opatchauto apply /u02/oracle/patches/35642822
```

OPatchauto session is initiated at Wed Dec 6 11:22:20 2023

System initialization log file is /u01/app/19.21.0/grid/cfgtoollogs/patchautodb/systemconfig2023-12-06_11-22-25AM.log.

Session log file is /u01/app/19.21.0/grid/cfgtoollogs/patchauto/patchauto2023-12-06_11-22-43AM.log
The id for this session is 8EMS

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.21.0/grid
Patch applicability verified successfully on home /u01/app/19.21.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Executing patch validation checks on home /u01/app/19.21.0/grid
Patch validation checks successfully completed on home /u01/app/19.21.0/grid

Executing patch validation checks on home /u01/app/oracle/product/19.21.0/dbhome_1
Patch validation checks successfully completed on home /u01/app/oracle/product/19.21.0/dbhome_1

Verifying SQL patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1
SQL patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Preparing to bring down database service on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully prepared home /u01/app/oracle/product/19.21.0/dbhome_1 to bring down database service

Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.21.0/grid
Prepatch operation log file location: /u01/app/grid/crsdata/gncashtl08922/crsconfig/crs_prepatch_apply_inplace_gncashtl08922_2023-12-06_11-24-25AM.log
CRS service brought down successfully on home /u01/app/19.21.0/grid

Performing prepatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Prepatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/oracle/product/19.21.0/dbhome_1
Binary patch applied successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Running rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully executed rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Performing postpatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Postpatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/19.21.0/grid
Binary patch applied successfully on home /u01/app/19.21.0/grid

Running rootadd_rdbms.sh on home /u01/app/19.21.0/grid
Successfully executed rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.21.0/grid
Postpatch operation log file location: /u01/app/grid/crsdata/gncashtl08922/crsconfig/crs_postpatch_apply_inplace_gncashtl08922_2023-12-06_11-38-11AM.log
CRS service started successfully on home /u01/app/19.21.0/grid

Preparing home /u01/app/oracle/product/19.21.0/dbhome_1 after database service restarted
No step execution required.....

Trying to apply SQL patch on home /u01/app/oracle/product/19.21.0/dbhome_1
SQL patch applied successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

OPatchAuto successful.

-----Summary-----

Patching is completed successfully. Please find the summary as follows:

Host:gncashtl08922

RAC Home:/u01/app/oracle/product/19.21.0/dbhome_1

Version:19.0.0.0.0

Summary:

==Following patches were SKIPPED:

Patch: /u02/oracle/patches/35642822/35652062

Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/35642822/33575402

Reason: This patch is not applicable to this specified target type - "rac_database"

Patch: /u02/oracle/patches/35642822/35553096

Reason: This patch is not applicable to this specified target type - "rac_database"

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35642822/35643107

Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-26-49AM_1.log

Patch: /u02/oracle/patches/35642822/35655527

Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-26-49AM_1.log

Host:gncashtl08922

CRS Home:/u01/app/19.21.0/grid

Version:19.0.0.0.0

Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35642822/33575402

Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-32-47AM_1.log

Patch: /u02/oracle/patches/35642822/35553096

Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-32-47AM_1.log

Patch: /u02/oracle/patches/35642822/35643107

Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-32-47AM_1.log

Patch: /u02/oracle/patches/35642822/35652062

Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-32-47AM_1.log

Patch: /u02/oracle/patches/35642822/35655527

Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_11-32-47AM_1.log

OPatchauto session completed at Wed Dec 6 11:57:09 2023

Time taken to complete the session 34 minutes, 45 seconds

[root@GNCASHTL08922 OPatch]#

JDK 19.21

Aplicando o JDK 19.21 no nó 01

[root@GNCASHTL08920 OPatch]# ./opatchauto apply /u02/oracle/patches/35638318

OPatchauto session is initiated at Wed Dec 6 15:22:50 2023

System initialization log file is /u01/app/19.21.0/grid/cfgtoollogs/patchautodb/systemconfig2023-12-06_03-22-52PM.log.

Session log file is /u01/app/19.21.0/grid/cfgtoollogs/patchauto/patchauto2023-12-06_03-23-11PM.log

The id for this session is MI 39

The ID for this session is 11255

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.21.0/grid

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1
Patch applicability verified successfully on home /u01/app/19.21.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Executing patch validation checks on home /u01/app/19.21.0/grid
Patch validation checks successfully completed on home /u01/app/19.21.0/grid

Executing patch validation checks on home /u01/app/oracle/product/19.21.0/dbhome_1
Patch validation checks successfully completed on home /u01/app/oracle/product/19.21.0/dbhome_1

Verifying SQL patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1
No sqlpatch prereq operations are required on the local node for this home
No step execution required.....

Preparing to bring down database service on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully prepared home /u01/app/oracle/product/19.21.0/dbhome_1 to bring down database service

Bringing down database service on home /u01/app/oracle/product/19.21.0/dbhome_1
No step execution required.....

Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.21.0/grid
Prepatch operation log file location: /u01/app/grid/crsdata/gncashtl08920/crsconfig/crs_prepatch_apply_inplace_gncashtl08920_2023-12-06_03-23-55PM.log
CRS service brought down successfully on home /u01/app/19.21.0/grid

Performing prepatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Prepatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/oracle/product/19.21.0/dbhome_1
Binary patch applied successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Running rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully executed rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Performing postpatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Postpatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/19.21.0/grid
Binary patch applied successfully on home /u01/app/19.21.0/grid

Running rootadd_rdbms.sh on home /u01/app/19.21.0/grid
Successfully executed rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.21.0/grid
Postpatch operation log file location: /u01/app/grid/crsdata/gncashtl08920/crsconfig/crs_postpatch_apply_inplace_gncashtl08920_2023-12-06_03-28-58PM.log
CRS service started successfully on home /u01/app/19.21.0/grid

Starting database service on home /u01/app/oracle/product/19.21.0/dbhome_1
No step execution required.....

Preparing home /u01/app/oracle/product/19.21.0/dbhome_1 after database service restarted
No step execution required.....

Trying to apply SQL patch on home /u01/app/oracle/product/19.21.0/dbhome_1
No SQL patch operations are required on local node for this home

OPatchAuto successful.

-----Summary-----

Patching is completed successfully. Please find the summary as follows:

Host:gncashtl08920
RAC Home:/u01/app/oracle/product/19.21.0/dbhome_1
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35638318
Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_15-27-03PM_1.log

Host:gncashtl08920
CRS Home:/u01/app/19.21.0/grid
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35638318
Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_15-28-03PM_1.log

OPatchauto session completed at Wed Dec 6 15:31:20 2023
Time taken to complete the session 8 minutes, 30 seconds

Aplicando o JDK 19.21 no nó 02

[root@GNCASHTL08921 OPatch]# ./opatchauto apply /u02/oracle/patches/35638318

OPatchauto session is initiated at Wed Dec 6 15:48:16 2023

System initialization log file is /u01/app/19.21.0/grid/cfgtoollogs/patchautodb/systemconfig2023-12-06_03-48-18PM.log.

Session log file is /u01/app/19.21.0/grid/cfgtoollogs/patchauto/patchauto2023-12-06_03-48-37PM.log
The id for this session is HIQN

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.21.0/grid
Patch applicability verified successfully on home /u01/app/19.21.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Executing patch validation checks on home /u01/app/19.21.0/grid
Patch validation checks successfully completed on home /u01/app/19.21.0/grid

Executing patch validation checks on home /u01/app/oracle/product/19.21.0/dbhome_1
Patch validation checks successfully completed on home /u01/app/oracle/product/19.21.0/dbhome_1

Verifying SQL patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1
No sqlpatch prereq operations are required on the local node for this home
No step execution required.....

Preparing to bring down database service on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully prepared home /u01/app/oracle/product/19.21.0/dbhome_1 to bring down database service

Bringing down database service on home /u01/app/oracle/product/19.21.0/dbhome_1
No step execution required.....

Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.21.0/grid
Prepatch operation log file location: /u01/app/grid/crsdata/gncashtl08921/crsconfig/crs_prepatch_apply_inplace_gncashtl08921_2023-12-06_03-49-18PM.log
CRS service brought down successfully on home /u01/app/19.21.0/grid

Performing prepatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Prepatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/oracle/product/19.21.0/dbhome_1
Binary patch applied successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Running rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1
Successfully executed rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Performing postpatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1
Postpatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/19.21.0/grid
Binary patch applied successfully on home /u01/app/19.21.0/grid

Running rootadd_rdbms.sh on home /u01/app/19.21.0/grid
Successfully executed rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.21.0/grid
Postpatch operation log file location: /u01/app/grid/crsdata/gncashtl08921/crsconfig/crs_postpatch_apply_inplace_gncashtl08921_2023-12-06_03-54-13PM.log
CRS service started successfully on home /u01/app/19.21.0/grid

Starting database service on home /u01/app/oracle/product/19.21.0/dbhome_1
No step execution required.....

Preparing home /u01/app/oracle/product/19.21.0/dbhome_1 after database service restarted
No step execution required.....

Trying to apply SQL patch on home /u01/app/oracle/product/19.21.0/dbhome_1
No SQL patch operations are required on local node for this home

OPatchAuto successful.

-----Summary-----

Patching is completed successfully. Please find the summary as follows:

Host:gncashtl08921
RAC Home:/u01/app/oracle/product/19.21.0/dbhome_1
Version:19.0.0.0.0
Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35638318

Patch: /u02/oracle/patches/35638318

Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_15-52-20PM_1.log

Host:gncashtl08921

CRS Home:/u01/app/19.21.0/grid

Version:19.0.0.0.0

Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35638318

Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_15-53-20PM_1.log

OPatchauto session completed at Wed Dec 6 15:56:29 2023

Time taken to complete the session 8 minutes, 13 seconds

Aplicando o JDK 19.21 no nó 03

[root@GNCASHTL08922 OPatch]# ./opatchauto apply /u02/oracle/patches/35638318

OPatchauto session is initiated at Wed Dec 6 15:56:43 2023

System initialization log file is /u01/app/19.21.0/grid/cfgtoollogs/patchautodb/systemconfig2023-12-06_03-56-45PM.log.

Session log file is /u01/app/19.21.0/grid/cfgtoollogs/patchauto/patchauto2023-12-06_03-57-05PM.log

The id for this session is M1JT

Executing OPatch prereq operations to verify patch applicability on home /u01/app/19.21.0/grid

Executing OPatch prereq operations to verify patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1

Patch applicability verified successfully on home /u01/app/19.21.0/grid

Patch applicability verified successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Executing patch validation checks on home /u01/app/19.21.0/grid

Patch validation checks successfully completed on home /u01/app/19.21.0/grid

Executing patch validation checks on home /u01/app/oracle/product/19.21.0/dbhome_1

Patch validation checks successfully completed on home /u01/app/oracle/product/19.21.0/dbhome_1

Verifying SQL patch applicability on home /u01/app/oracle/product/19.21.0/dbhome_1

No sqlpatch prereq operations are required on the local node for this home

No step execution required.....

Preparing to bring down database service on home /u01/app/oracle/product/19.21.0/dbhome_1

Successfully prepared home /u01/app/oracle/product/19.21.0/dbhome_1 to bring down database service

Bringing down database service on home /u01/app/oracle/product/19.21.0/dbhome_1

No step execution required.....

Performing prepatch operations on CRS - bringing down CRS service on home /u01/app/19.21.0/grid

Prepatch operation log file location: /u01/app/grid/crsdata/gncashtl08922/crsconfig/crs_prepatch_apply_inplace_gncashtl08922_2023-12-06_03-57-48PM.log

CRS service brought down successfully on home /u01/app/19.21.0/grid

Performing prepatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1

Prepatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/oracle/product/19.21.0/dbhome_1

Binary patch applied successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Running rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Successfully executed rootadd_rdbms.sh on home /u01/app/oracle/product/19.21.0/dbhome_1

Performing postpatch operation on home /u01/app/oracle/product/19.21.0/dbhome_1

Postpatch operation completed successfully on home /u01/app/oracle/product/19.21.0/dbhome_1

Start applying binary patch on home /u01/app/19.21.0/grid

Binary patch applied successfully on home /u01/app/19.21.0/grid

Running rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Successfully executed rootadd_rdbms.sh on home /u01/app/19.21.0/grid

Performing postpatch operations on CRS - starting CRS service on home /u01/app/19.21.0/grid

Postpatch operation log file location: /u01/app/grid/crsdata/gncashtl08922/crsconfig/crs_postpatch_apply_inplace_gncashtl08922_2023-12-06_04-02-39PM.log

CRS service started successfully on home /u01/app/19.21.0/grid

Starting database service on home /u01/app/oracle/product/19.21.0/dbhome_1

No step execution required.....

Preparing home /u01/app/oracle/product/19.21.0/dbhome_1 after database service restarted

No step execution required.....

Trying to apply SQL patch on home /u01/app/oracle/product/19.21.0/dbhome_1

No SQL patch operations are required on local node for this home

OPatchAuto successful.

-----Summary-----

Patching is completed successfully. Please find the summary as follows:

Host:gncashtl08922

RAC Home:/u01/app/oracle/product/19.21.0/dbhome_1

Version:19.0.0.0.0

Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35638318

Log: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/patchauto/core/patch/patch2023-12-06_16-00-46PM_1.log

Host:gncashtl08922

CRS Home:/u01/app/19.21.0/grid

Version:19.0.0.0.0

Summary:

==Following patches were SUCCESSFULLY applied:

Patch: /u02/oracle/patches/35638318

Log: /u01/app/19.21.0/grid/cfgtoollogs/patchauto/core/patch/patch2023-12-06_16-01-45PM_1.log

OPatchauto session completed at Wed Dec 6 16:05:04 2023

Time taken to complete the session 8 minutes, 21 seconds

OJVM 19.21

```

[oracle@GNCASHTL08920 ~]$ . db_profile
[oracle@GNCASHTL08920 ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Dec 6 16:09:00 2023
Version 19.21.0.0.0

Copyright (c) 1982, 2022, Oracle. All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0


SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0
[oracle@GNCASHTL08920 ~]$ cd $ORACLE_HOME
[oracle@GNCASHTL08920 dbhome_1]$ cd OPatch/
[oracle@GNCASHTL08920 OPatch]$ ./opatch apply /u02/oracle/patches/35648110
Oracle Interim Patch Installer version 12.2.0.1.40
Copyright (c) 2023, Oracle Corporation. All rights reserved.


Oracle Home      : /u01/app/oracle/product/19.21.0/dbhome_1
Central Inventory : /u01/app/oraInventory
   from           : /u01/app/oracle/product/19.21.0/dbhome_1/oraInst.loc
OPatch version    : 12.2.0.1.40
OUI version       : 12.2.0.7.0
Log file location : /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/opatch/opatch2023-12-06_16-10-11PM_1.log


Verifying environment and performing prerequisite checks...
OPatch continues with these patches: 35648110


Do you want to proceed? [y|n]
y
User Responded with: Y
All checks passed.


Please shutdown Oracle instances running out of this ORACLE_HOME on the local system.
(Oracle Home = '/u01/app/oracle/product/19.21.0/dbhome_1')


Is the local system ready for patching? [y|n]
y
User Responded with: Y
Backing up files...
Applying interim patch '35648110' to OH '/u01/app/oracle/product/19.21.0/dbhome_1'


Patching component oracle.javavm.server, 19.0.0.0.0...

Patching component oracle.javavm.server.core, 19.0.0.0.0...

Patching component oracle.rdbms.dbscripts, 19.0.0.0.0...

Patching component oracle.rdbms, 19.0.0.0.0...

Patching component oracle.javavm.client, 19.0.0.0.0...
Patch 35648110 successfully applied.
Log file location: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/opatch/opatch2023-12-06_16-10-11PM_1.log


OPatch succeeded.
[oracle@GNCASHTL08920 OPatch]$ sqlplus / as sysdba

```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed Dec 6 16:13:06 2023  
Version 19.21.0.0.0
```

Copyright (c) 1982, 2022, Oracle. All rights reserved.

Connected to an idle instance.

```
SQL> startup  
ORACLE instance started.
```

```
Total System Global Area 4949276544 bytes  
Fixed Size      8935296 bytes  
Variable Size   1140850688 bytes  
Database Buffers 3791650816 bytes  
Redo Buffers    7839744 bytes  
Database mounted.  
Database opened.  
SQL>
```

Aplicando o OJVM 19.21 no nó 02

```
[oracle@GNCASHTL08921 ~]$ . db_profile  
[oracle@GNCASHTL08921 ~]$ cd $ORACLE_HOME  
[oracle@GNCASHTL08921 dbhome_1]$ cd OPatch/  
[oracle@GNCASHTL08921 OPatch]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed Dec 6 16:14:13 2023  
Version 19.21.0.0.0
```

Copyright (c) 1982, 2022, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0

```
SQL> shutdown immediate;  
Database closed.  
Database dismounted.  
ORACLE instance shut down.  
SQL> exit  
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.21.0.0.0  
[oracle@GNCASHTL08921 OPatch]$ ./opatch apply /u02/oracle/patches/35648110  
Oracle Interim Patch Installer version 12.2.0.1.40  
Copyright (c) 2023, Oracle Corporation. All rights reserved.
```

```
Oracle Home      : /u01/app/oracle/product/19.21.0/dbhome_1  
Central Inventory : /u01/app/oraInventory  
   from           : /u01/app/oracle/product/19.21.0/dbhome_1/oraInst.loc  
OPatch version   : 12.2.0.1.40  
OUI version      : 12.2.0.7.0  
Log file location : /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/opatch/opatch2023-12-06_16-15-40PM_1.log
```

```
Verifying environment and performing prerequisite checks...  
OPatch continues with these patches: 35648110
```

```
Do you want to proceed? [y|n]  
y  
User Responded with: Y  
All checks passed.
```

```
Please shutdown Oracle instances running out of this ORACLE_HOME on the local system.  
(Oracle Home = '/u01/app/oracle/product/19.21.0/dbhome_1')
```

```
Is the local system ready for patching? [y|n]  
y  
User Responded with: Y
```

```

Backing up files...
Applying interim patch '35648110' to OH '/u01/app/oracle/product/19.21.0/dbhome_1'

Patching component oracle.javavm.server, 19.0.0.0.0...

Patching component oracle.javavm.server.core, 19.0.0.0.0...

Patching component oracle.rdbms.dbscripts, 19.0.0.0.0...

Patching component oracle.rdbms, 19.0.0.0.0...

Patching component oracle.javavm.client, 19.0.0.0.0...
Patch 35648110 successfully applied.
Log file location: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/opatch/opatch2023-12-06_16-15-40PM_1.log

OPatch succeeded.
[oracle@GNCASHTL08921 OPatch]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Dec 6 16:20:14 2023
Version 19.21.0.0.0

Copyright (c) 1982, 2022, Oracle. All rights reserved.

sConnected to an idle instance.

SQL> startup
ORACLE instance started.

Total System Global Area 4949276544 bytes
Fixed Size 8935296 bytes
Variable Size 1157627904 bytes
Database Buffers 3774873600 bytes
Redo Buffers 7839744 bytes
Database mounted.
Database opened.
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0
[oracle@GNCASHTL08921 OPatch]$

```

Aplicando o OJVM 19.21 no nó 03

```

[oracle@GNCASHTL08922 ~]$ . db_profile
[oracle@GNCASHTL08922 ~]$ cd $ORACLE_HOME
[oracle@GNCASHTL08922 dbhome_1]$ cd OPatch/
[oracle@GNCASHTL08922 OPatch]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Dec 6 16:20:40 2023
Version 19.21.0.0.0

Copyright (c) 1982, 2022, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0
[oracle@GNCASHTL08922 OPatch]$ ./opatch apply /u02/oracle/patches/35648110
Oracle Interim Patch Installer version 12.2.0.1.40
Copyright (c) 2023, Oracle Corporation. All rights reserved.

Oracle Home      : /u01/app/oracle/product/19.21.0/dbhome_1

```

```
Central Inventory : /u01/app/oraInventory
  from           : /u01/app/oracle/product/19.21.0/dbhome_1/oraInst.loc
OPatch version   : 12.2.0.1.40
OUI version      : 12.2.0.7.0
Log file location : /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/opatch/opatch2023-12-06_16-21-24PM_1.log
```

```
Verifying environment and performing prerequisite checks...
OPatch continues with these patches: 35648110
```

```
Do you want to proceed? [y|n]
y
User Responded with: Y
All checks passed.
```

```
Please shutdown Oracle instances running out of this ORACLE_HOME on the local system.
(Oracle Home = '/u01/app/oracle/product/19.21.0/dbhome_1')
```

```
Is the local system ready for patching? [y|n]
y
User Responded with: Y
Backing up files...
Applying interim patch '35648110' to OH '/u01/app/oracle/product/19.21.0/dbhome_1'
```

```
Patching component oracle.javavm.server, 19.0.0.0.0...
```

```
Patching component oracle.javavm.server.core, 19.0.0.0.0...
```

```
Patching component oracle.rdbms.dbscripts, 19.0.0.0.0...
```

```
Patching component oracle.rdbms, 19.0.0.0.0...
```

```
Patching component oracle.javavm.client, 19.0.0.0.0...
```

```
Patch 35648110 successfully applied.
```

```
Log file location: /u01/app/oracle/product/19.21.0/dbhome_1/cfgtoollogs/opatch/opatch2023-12-06_16-21-24PM_1.log
```

```
OPatch succeeded.
```

```
[oracle@GNCASHTL08922 OPatch]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed Dec 6 16:23:06 2023
Version 19.21.0.0.0
```

```
Copyright (c) 1982, 2022, Oracle. All rights reserved.
```

```
stConnected to an idle instance.
```

```
SQL> artup
ORACLE instance started.
```

```
Total System Global Area 4949276544 bytes
Fixed Size                  8935296 bytes
Variable Size              1426063360 bytes
Database Buffers           3506438144 bytes
Redo Buffers                7839744 bytes
Database mounted.
Database opened.
SQL>
```

Datapatch OJVM 19.21


```
[oracle@GNCASHTL08922 OPatch]$ ./datapatch -verbose
SQL Patching tool version 19.21.0.0.0 Production on Wed Dec 6 16:24:04 2023
Copyright (c) 2012, 2023, Oracle. All rights reserved.

Log file for this invocation: /u01/app/oracle/cfgtoollogs/sqlpatch/sqlpatch_1502127_2023_12_06_16_24_04/sqlpatch_invocation.log

Connecting to database...OK
Gathering database info...done
Bootstrapping registry and package to current versions...done
Determining current state...done

Current state of interim SQL patches:
Interim patch 35648110 (OJVM RELEASE UPDATE: 19.21.0.0.231017 (35648110)):
  Binary registry: Installed
  SQL registry: Not installed

Current state of release update SQL patches:
Binary registry:
  19.21.0.0.0 Release_Update 230930151951: Installed
SQL registry:
  Applied 19.21.0.0.0 Release_Update 230930151951 successfully on 06-DEC-23 11.57.06.638074 AM

Adding patches to installation queue and performing prereq checks...done
Installation queue:
  No interim patches need to be rolled back
  No release update patches need to be installed
  The following interim patches will be applied:
    35648110 (OJVM RELEASE UPDATE: 19.21.0.0.231017 (35648110))

Installing patches...
Patch installation complete. Total patches installed: 1

Validating logfiles...done
Patch 35648110 apply: SUCCESS
  logfile: /u01/app/oracle/cfgtoollogs/sqlpatch/35648110/25365038/35648110_apply_TAPONPHO_2023Dec06_16_24_36.log (no errors)
SQL Patching tool complete on Wed Dec 6 16:24:37 2023
[oracle@GNCASHTL08922 OPatch]$
```

Evidências banco:

```
SQL> select instance_name, host_name, version, status, open_mode, database_role, switchover_status from gv$instance, v$database;
```

INSTANCE_NAME	HOST_NAME	VERSION	STATUS	OPEN_MODE	DATABASE_ROLE	SWITCHOVER_STATUS
TAPONPHO3	GNCASHTL08922	19.0.0.0.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED
TAPONPHO2	GNCASHTL08921	19.0.0.0.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED
TAPONPHO1	GNCASHTL08920	19.0.0.0.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED

```
SQL> select comp_name, version, status, modified from dba_registry;
```

COMP_NAME	VERSION	STATUS	MODIFIED
Oracle Database Catalog Views	19.0.0.0.0	VALID	06-DEC-2023 10:25:46
Oracle Database Packages and Types	19.0.0.0.0	VALID	06-DEC-2023 10:25:46
Oracle Real Application Clusters	19.0.0.0.0	VALID	06-DEC-2023 10:25:48
JServer JAVA Virtual Machine	19.0.0.0.0	VALID	06-DEC-2023 16:24:37
Oracle XDK	19.0.0.0.0	VALID	06-DEC-2023 10:25:48
Oracle Database Java Packages	19.0.0.0.0	VALID	06-DEC-2023 10:25:48
OLAP Analytic Workspace	19.0.0.0.0	VALID	06-DEC-2023 10:25:49
Oracle XML Database	19.0.0.0.0	VALID	06-DEC-2023 10:25:47
Oracle Workspace Manager	19.0.0.0.0	VALID	06-DEC-2023 10:25:48
Oracle Text	19.0.0.0.0	VALID	06-DEC-2023 10:25:48
Oracle Multimedia	19.0.0.0.0	VALID	06-DEC-2023 10:25:48
Spatial	19.0.0.0.0	VALID	06-DEC-2023 16:35:01
Oracle OLAP API	19.0.0.0.0	VALID	06-DEC-2023 11:56:21
Oracle Label Security	19.0.0.0.0	VALID	06-DEC-2023 10:25:49
Oracle Database Vault	19.0.0.0.0	VALID	06-DEC-2023 10:25:49

15 rows selected.

```
SQL> select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;
```

PATCH_ID	ACTION	STATUS	ACTION_TIME	DESCRIPTION
29517242	APPLY	SUCCESS	06-DEC-23 10.23.43.747454 AM	Database Release Update : 19.3.0.0.190416 (29517242)
35643107	APPLY	SUCCESS	06-DEC-23 11.57.06.638074 AM	Database Release Update : 19.21.0.0.231017 (35643107)
35648110	APPLY	SUCCESS	06-DEC-23 04.24.37.914473 PM	OJVM RELEASE UPDATE: 19.21.0.0.231017 (35648110)

```
SQL>
```

Evidencias Grid Infrastructure:

```
[root@GNCASHTL08922 OPatch]# crsctl status res -t
```

Name	Target	State	Server	State details
Local Resources				
ora.LISTENER.lsnr				
	ONLINE	ONLINE	gncashtl08920	STABLE
	ONLINE	ONLINE	gncashtl08921	STABLE
	ONLINE	ONLINE	gncashtl08922	STABLE
ora.chad				
	ONLINE	ONLINE	gncashtl08920	STABLE
	ONLINE	ONLINE	gncashtl08921	STABLE
	ONLINE	ONLINE	gncashtl08922	STABLE
ora.net1.network				
	ONLINE	ONLINE	gncashtl08920	STABLE
	ONLINE	ONLINE	gncashtl08921	STABLE
	ONLINE	ONLINE	gncashtl08922	STABLE
ora.ons				
	ONLINE	ONLINE	gncashtl08920	STABLE
	ONLINE	ONLINE	gncashtl08921	STABLE
	ONLINE	ONLINE	gncashtl08922	STABLE

Cluster Resources

```
ora.ARCH_TAPONPHONE.dg(ora.asmgroup)
1 ONLINE ONLINE gncashtl08920 STABLE
```

```

1    ONLINE  ONLINE  gncashtl08920    STABLE
2    ONLINE  ONLINE  gncashtl08921    STABLE
3    ONLINE  ONLINE  gncashtl08922    STABLE
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
1    ONLINE  ONLINE  gncashtl08920    STABLE
2    ONLINE  ONLINE  gncashtl08921    STABLE
3    ONLINE  ONLINE  gncashtl08922    STABLE
ora.DATA_AUX_TAPONPHONE.dg(ora.asmgroup)
1    ONLINE  ONLINE  gncashtl08920    STABLE
2    ONLINE  ONLINE  gncashtl08921    STABLE
3    ONLINE  ONLINE  gncashtl08922    STABLE
ora.DATA_TAPONPHONE.dg(ora.asmgroup)
1    ONLINE  ONLINE  gncashtl08920    STABLE
2    ONLINE  ONLINE  gncashtl08921    STABLE
3    ONLINE  ONLINE  gncashtl08922    STABLE
ora.LISTENER_SCAN1.lsnr
1    ONLINE  ONLINE  gncashtl08921    STABLE
ora.LISTENER_SCAN2.lsnr
1    ONLINE  ONLINE  gncashtl08922    STABLE
ora.LISTENER_SCAN3.lsnr
1    ONLINE  ONLINE  gncashtl08920    STABLE
ora.asm(ora.asmgroup)
1    ONLINE  ONLINE  gncashtl08920    Started,STABLE
2    ONLINE  ONLINE  gncashtl08921    Started,STABLE
3    ONLINE  ONLINE  gncashtl08922    Started,STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
1    ONLINE  ONLINE  gncashtl08920    STABLE
2    ONLINE  ONLINE  gncashtl08921    STABLE
3    ONLINE  ONLINE  gncashtl08922    STABLE
ora.cvu
1    ONLINE  ONLINE  gncashtl08921    STABLE
ora.gncashtl08920.vip
1    ONLINE  ONLINE  gncashtl08920    STABLE
ora.gncashtl08921.vip
1    ONLINE  ONLINE  gncashtl08921    STABLE
ora.gncashtl08922.vip
1    ONLINE  ONLINE  gncashtl08922    STABLE
ora.qosmserver
1    ONLINE  ONLINE  gncashtl08920    STABLE
ora.scan1.vip
1    ONLINE  ONLINE  gncashtl08921    STABLE
ora.scan2.vip
1    ONLINE  ONLINE  gncashtl08922    STABLE
ora.scan3.vip
1    ONLINE  ONLINE  gncashtl08920    STABLE
ora.taponphone.db
1    ONLINE  ONLINE  gncashtl08920    Open,HOME=/u01/app/o
                   racle/product/19.21.
                   0/dbhome_1,STABLE
2    ONLINE  ONLINE  gncashtl08921    Open,HOME=/u01/app/o
                   racle/product/19.21.
                   0/dbhome_1,STABLE
3    ONLINE  ONLINE  gncashtl08922    Open,HOME=/u01/app/o
                   racle/product/19.21.
                   0/dbhome_1,STABLE

```

```
[root@GNCASHTL08922 OPatch]#
```

Banco em Archivelog

Colocando o banco em Archivelog.

Baixando o banco:

```
[oracle@GNCASHTL08920 OPatch]$ srvctl stop database -d TAPONPHONE
[oracle@GNCASHTL08920 OPatch]$ ps -ef |grep pmon
grid   1343614      1  0 Dec06 ?        00:00:03 asm_pmon_+ASM1
oracle 2180295 1406202  0 11:36 pts/0    00:00:00 grep --color=auto pmon

[oracle@GNCASHTL08921 OPatch]$ ps -ef |grep pmon
grid   1299484      1  0 Dec06 ?        00:00:02 asm_pmon_+ASM2
oracle 2107408 1342697  0 11:36 pts/0    00:00:00 grep --color=auto pmon

[oracle@GNCASHTL08922 ~]$ ps -ef |grep pmon
grid   1482838      1  0 Dec06 ?        00:00:02 asm_pmon_+ASM3
oracle 2198341 2193262  0 11:36 pts/0    00:00:00 grep --color=auto pmon
```

Subindo em mount:

```
[oracle@GNCASHTL08920 OPatch]$ srvctl start database -d TAPONPHONE -o mount
[oracle@GNCASHTL08920 OPatch]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Dec 7 11:40:27 2023
Version 19.21.0.0.0

Copyright (c) 1982, 2022, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0

SQL> alter database archivelog;

Database altered.

SQL>
```

Restart e configuração do destination:

```
[oracle@GNCASHTL08920 OPatch]$ srvctl stop database -d TAPONPHONE
[oracle@GNCASHTL08920 OPatch]$ srvctl start database -d TAPONPHONE
[oracle@GNCASHTL08920 OPatch]$ sqlplus / as sysdba
```

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Dec 7 11:43:14 2023
Version 19.21.0.0.0

Copyright (c) 1982, 2022, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.21.0.0.0

```
SQL> set lines 300
SELECT name, type, total_mb, free_mb, required_mirror_free_mb,
usable_file_mb FROM V$ASM_DISKGROUP;SQL> 2
```

NAME	TYPE	TOTAL_MB	FREE_MB	REQUIRED_MIRROR_FREE_MB	USABLE_FILE_MB
DATA_TAPONPHONE	EXTERN	148476	143692	0	143692
DATA_AUX_TAPONPHONE	EXTERN	37884	37672	0	37672
ARCH_TAPONPHONE	EXTERN	148476	148264	0	148264

```
SQL> alter system set log_archive_dest_1='LOCATION=+ARCH_TAPONPHONE' scope=both sid='*';
```

System altered.

```
SQL> archive log list
Database log mode          Archive Mode
Automatic archival         Enabled
Archive destination        +ARCH_TAPONPHONE
Oldest online log sequence 8
Next log sequence to archive 9
Current log sequence       9
SQL>
```

BookDBA

Aplicados os 4 scripts do book dba e configurados os scripts do contrab oracle e grid nos 3 nós:
NOTA: exemplo dos scripts em 1 nó:

```
[root@GNCASHTL08920 OPatch]# chmod 775 /u02/scripts/delete_logs_oracle.sh
[root@GNCASHTL08920 OPatch]# chown oracle:oinstall /u02/scripts/delete_logs_oracle.sh
[root@GNCASHTL08920 OPatch]# cat /u02/scripts/delete_logs_oracle.sh
#!/bin/bash
find /u01/app/oracle/diag/rdbms/taponphone/TAPONPHO1/trace/* -name "*.tr*" -mtime +7 -exec rm -rf {} \;
find /u01/app/oracle/diag/rdbms/taponphone/TAPONPHO1/cdump/* -name "*.tr*" -mtime +7 -exec rm -rf {} \;
find /u01/app/oracle/product/19.21.0/dbhome_1/rdbms/audit/* -mtime +7 -exec rm -rf {} \;
find /u01/app/oracle/admin/TAPONPHONE/adump/* -name "*.aud" -mtime +7 -exec rm -rf {} \;

[root@GNCASHTL08920 OPatch]# chown grid:oinstall /u02/scripts/delete_logs_grid.sh
[root@GNCASHTL08920 OPatch]# chmod 775 /u02/scripts/delete_logs_grid.sh
[root@GNCASHTL08920 OPatch]# cat /u02/scripts/delete_logs_grid.sh
#!/bin/bash
find /u01/app/grid/diag/asm/+asm/+ASM1/incident/* -mtime +7 -exec rm -rf {} \;
find /u01/app/grid/diag/asm/+asm/+ASM1/alert/* -mtime +7 -exec rm -rf {} \;
find /u01/app/grid/diag/asm/+asm/+ASM1/trace/*.tr* -mtime +7 -exec rm -rf {} \;
find /u01/app/19.21.0/grid/rdbms/audit/ -name "*.aud" -mtime +7 -exec rm -rf {} \;
```